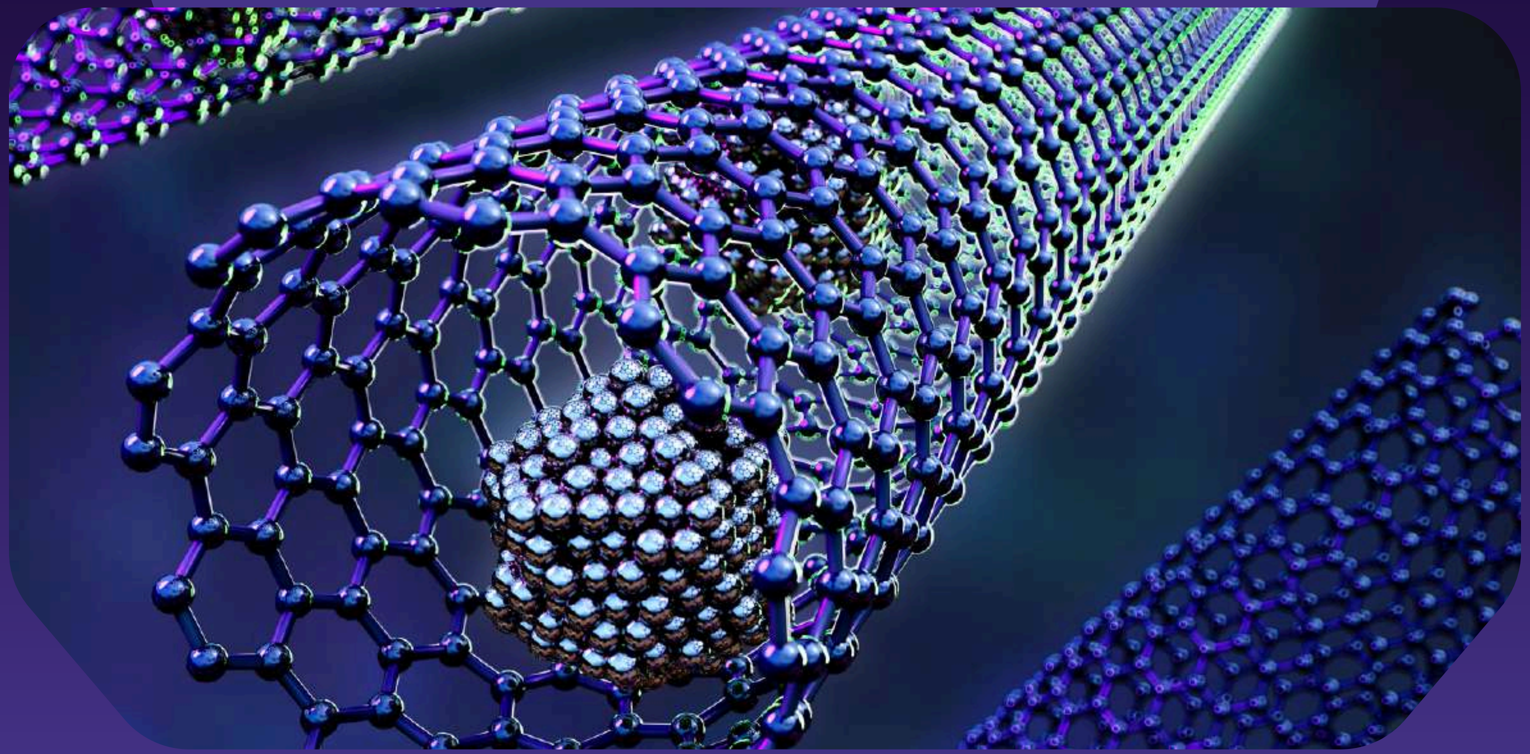
SOUTHERN LUZON STATE UNIVERSITY
GRADUATE SCHOOL
SCI 513 Advanced Topics in Chemistry



Engineering at the Scale of Atoms and Molecules

Presented by
MARVIN GAMBAN
JEDIDIA MAE LINASAN
RICA JOY OCAMPO

NANOCHEMISTRY

LEARNING OBJECTIVES

- Define nanochemistry and describe the unique properties and characteristics of nanomaterials.
- Differentiate the top-down and bottom-up approaches and identify various types of nanomaterials.
- Examine how nanochemistry is applied in medicine, environment, energy, and technology, including its benefits and risks.
- Analyze information from research journals and educational videos related to nanochemistry.
- Connect nanochemistry concepts to fundamental science competencies and real-world applications.

KEY TERMS

NANOCHEMISTRY

THE BRANCH OF CHEMISTRY THAT STUDIES MATERIALS AT THE NANOSCALE (1–100 NANOMETERS).

NANOTECHNOLOGY

THE APPLICATION OF NANOCHEMISTRY IN CREATING MATERIALS AND DEVICES AT THE NANOSCALE.

NANOPARTICLES

EXTREMELY SMALL PARTICLES WITH DIMENSIONS MEASURED IN NANOMETERS.

SURFACE AREA-TO-VOLUME RATIO

A PROPERTY THAT INCREASES AS PARTICLE SIZE DECREASES, AFFECTING REACTIVITY.

QUANTUM EFFECTS

CHANGES IN MATERIAL BEHAVIOR AT THE NANOSCALE DUE TO ATOMIC AND SUBATOMIC INTERACTIONS.

TOP-DOWN APPROACH

A METHOD OF CREATING NANOMATERIALS BY BREAKING DOWN BULK MATERIALS INTO SMALLER PIECES.

BOTTOM-UP APPROACH

A METHOD OF BUILDING NANOMATERIALS ATOM-BY-ATOM OR MOLECULE-BY-MOLECULE.

CARBON NANOTUBES

CYLINDRICAL NANOSTRUCTURES MADE OF CARBON WITH HIGH STRENGTH AND CONDUCTIVITY.

QUANTUM DOTS

SEMICONDUCTOR NANOPARTICLES THAT EMIT LIGHT AND ARE USED IN IMAGING AND ELECTRONICS.

SELF-ASSEMBLY

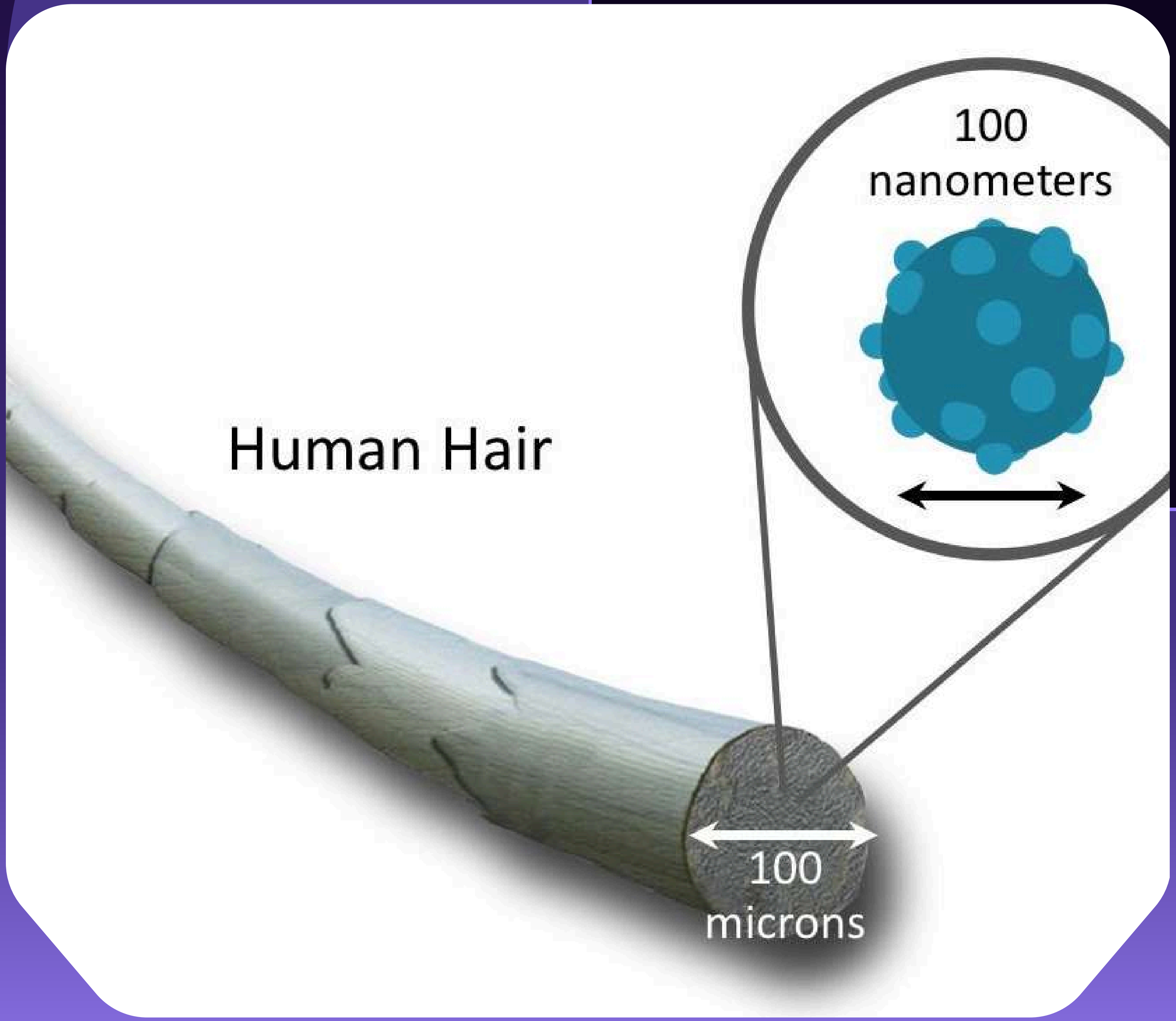
THE PROCESS WHERE MOLECULES AUTOMATICALLY ORGANIZE INTO STRUCTURED ARRANGEMENTS WITHOUT EXTERNAL GUIDANCE.

WHAT IS NANOCHEMISTRY?

- Nanochemistry is a branch of chemistry that focuses on the study and manipulation of materials at the nanoscale (1–100 nanometers).
- It involves designing, synthesizing, and analyzing materials with extremely small dimensions.

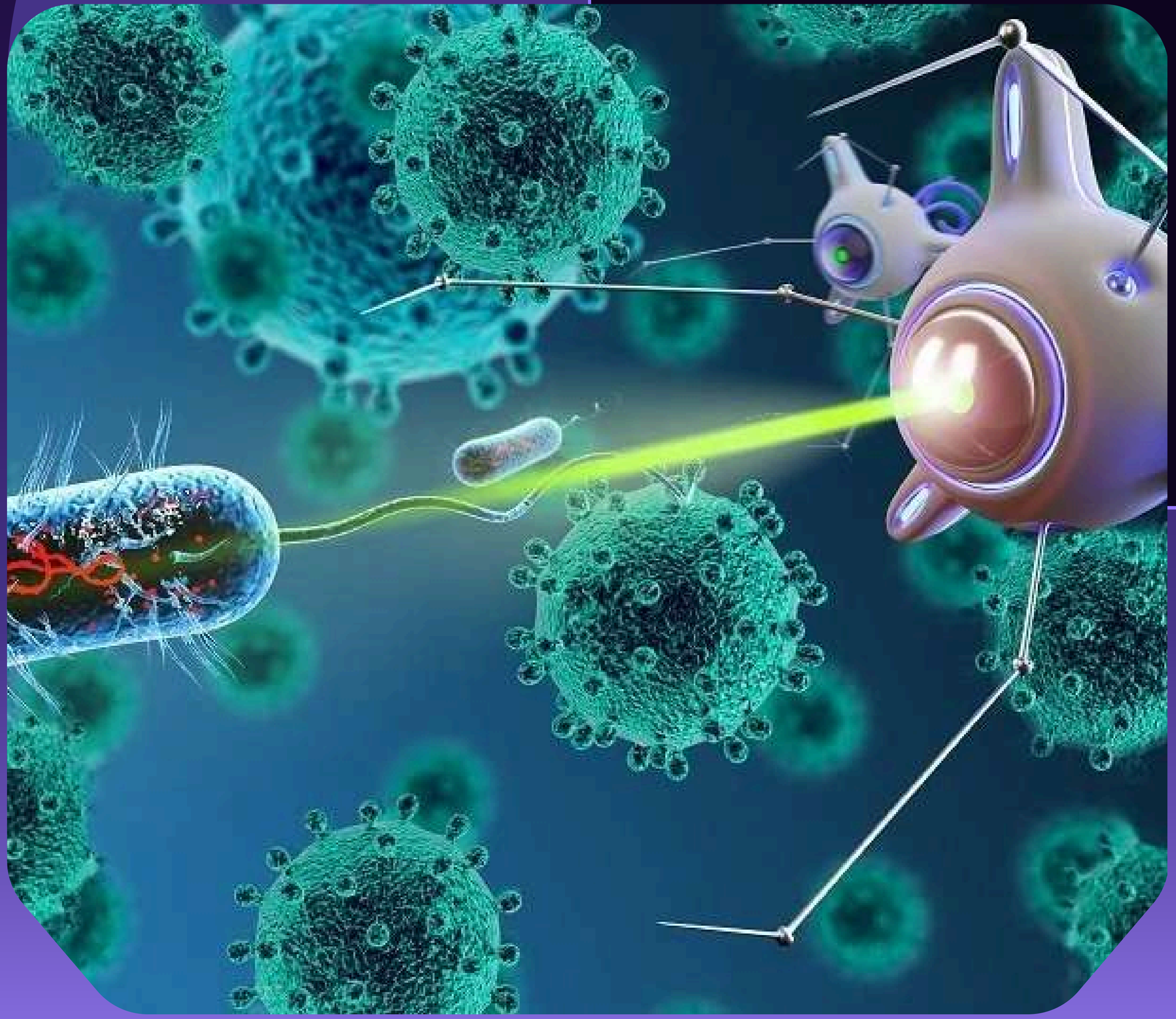
NOTE:

- 1 NANOMETER = 1 BILLIONTH OF A METER
- COMPARE: HUMAN HAIR ≈ 80,000–100,000 NM WIDE



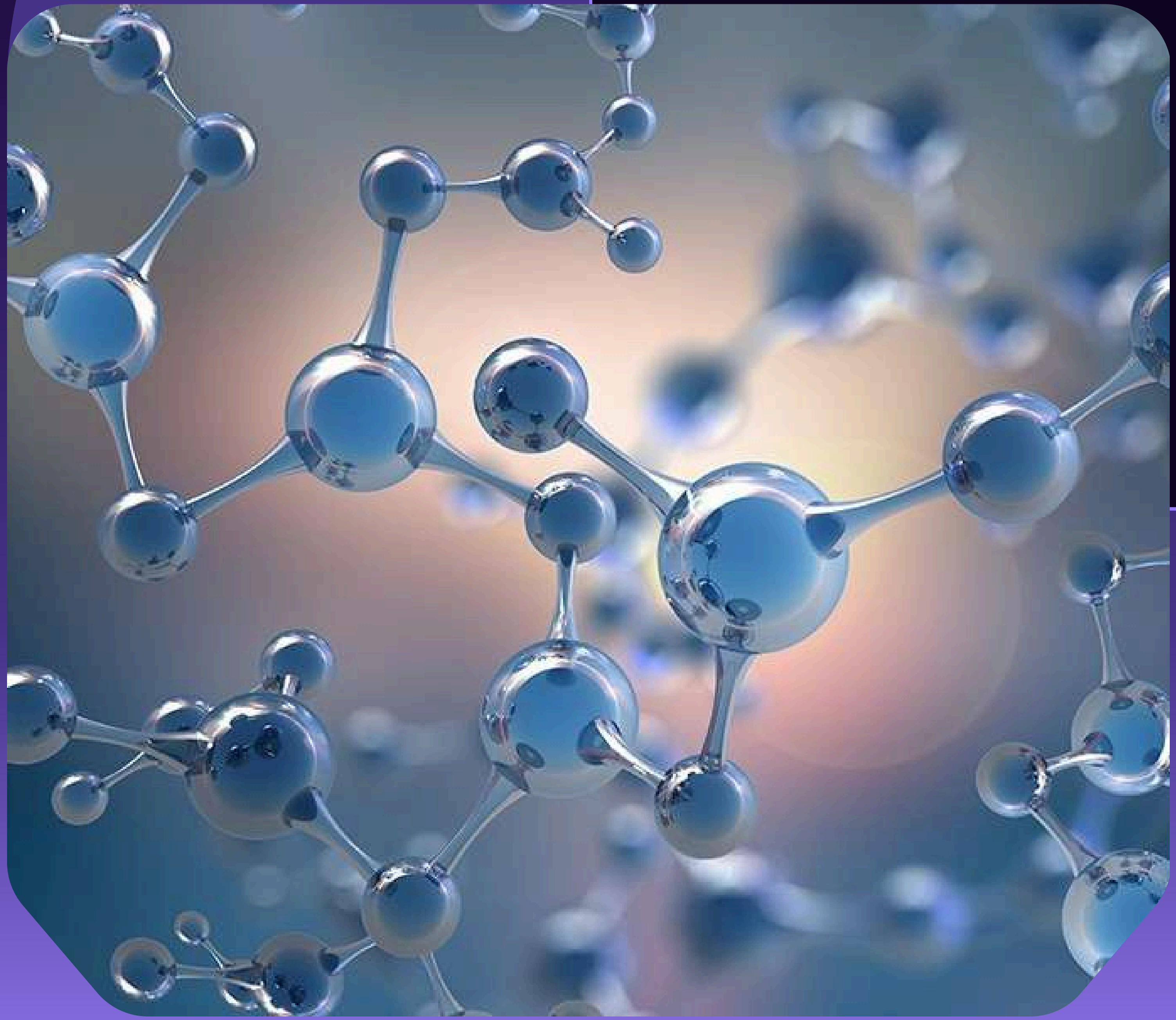
IMPORTANCE OF NANOCHEMISTRY

- Enables development of advanced materials with improved properties
- Plays a key role in modern innovations such as:
  - Medicine (targeted drug delivery)
  - Electronics (smaller, faster devices)
  - Environmental solutions (pollution control)



WHAT MAKES NANOCHEMISTRY UNIQUE?

- Materials behave differently at the nanoscale compared to bulk materials
- Two main reasons:
  - Increased surface area-to-volume ratio
  - Presence of quantum effects

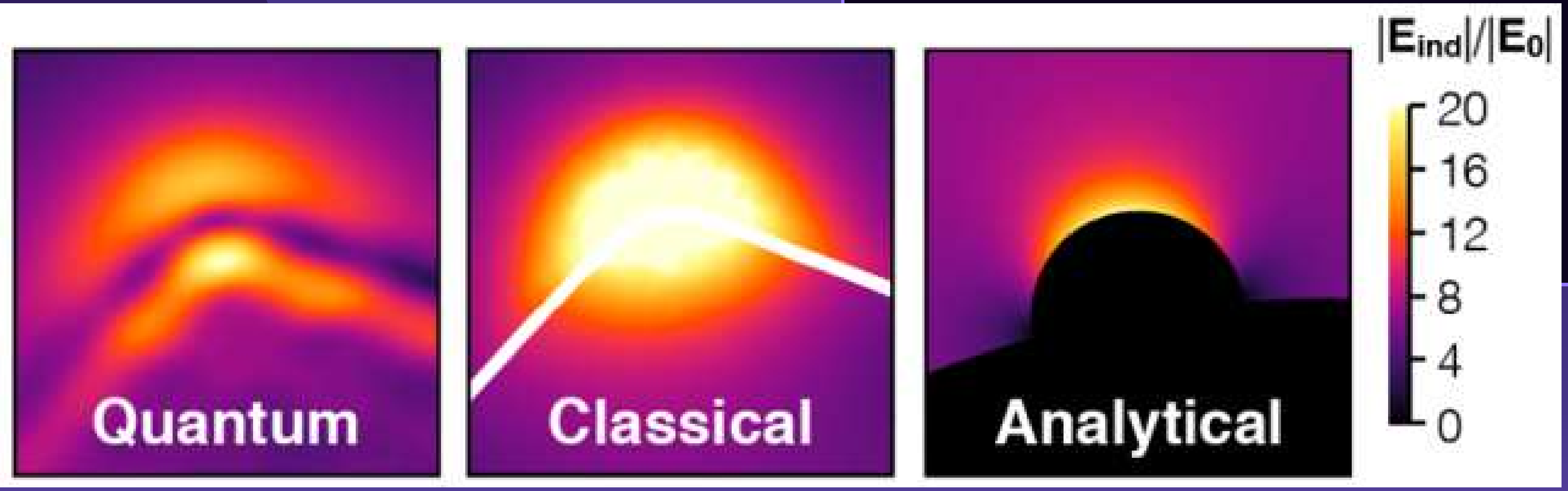


PROPERTY 1: SURFACE AREA-TO-VOLUME RATIO

- As particle size decreases, surface area increases relative to volume
- More atoms are exposed on the surface
- Effects:
    - Higher reactivity
    - Faster chemical reactions
    - Improved catalytic properties
- Example:
    - Nanoparticles react faster than larger particles of the same material

PROPERTY 2: QUANTUM EFFECTS

- At the nanoscale, classical physics no longer fully explains behavior
- Electrons behave differently, affecting:
    - Color
    - Electrical conductivity
    - Optical properties
- Example:
    - Gold nanoparticles can appear red or purple instead of metallic gold



ENHANCED CHEMICAL REACTIVITY

- Nanomaterials are more reactive due to:
  - Larger exposed surface area
  - More active sites for reactions
- Applications:
  - Catalysts in chemical reactions
  - Environmental cleanup (breaking down pollutants)

UNIQUE PHYSICAL PROPERTIES

- Nanomaterials may have:
  - Increased strength and durability
  - Higher electrical conductivity
  - Improved thermal properties
- Examples:
  - Carbon nanotubes are stronger than steel but very lightweight
  - Nanomaterials in electronics improve performance

Characteristic	Description	Key Effects / Examples
Nanoscale Size (1–100 nm)	Materials are extremely small and cannot be seen with the naked eye.	Requires special tools (e.g., electron microscope) to observe.
High Surface Area-to-Volume Ratio	Smaller particles have more exposed surface atoms.	Increased reactivity; faster chemical reactions; effective catalysts.
Quantum Effects	Electron behavior changes at very small scales.	Changes in color, conductivity, and optical properties (e.g., gold nanoparticles appear red/purple).
Enhanced Chemical Reactivity	More active sites allow more reactions to occur.	Used in catalysis and pollution control.
Unique Physical Properties	Materials exhibit improved strength, conductivity, and thermal properties.	Carbon nanotubes are strong yet lightweight; improved electronics.
Size-Dependent Properties	Properties change depending on size and shape.	Different sizes can produce different colors or reactivity.
Self-Assembly Capability	Molecules can organize themselves into structured arrangements.	Formation of complex nanostructures without external control.
Interdisciplinary Nature	Combines multiple scientific fields.	Applied in medicine, engineering, environmental science, and electronics.

THERMODYNAMIC PROPERTIES

- Nanomaterials have higher surface energy due to more exposed atoms
- They are less thermodynamically stable than bulk materials
- They often have lower melting points
- Key Idea:
  - More surface atoms = higher energy and reactivity
- Examples/Applications:
  - Gold nanoparticles melt at lower temperatures
  - Used in catalysts for faster reactions

MECHANICAL PROPERTIES

- Nanomaterials show increased strength and hardness
- Some are lightweight yet extremely durable
- Can also show flexibility depending on structure
- Key Idea:
  - Stronger materials can be made using nanoscale structures
- Examples/Applications:
  - Carbon nanotubes are stronger than steel
  - Used in aerospace, sports equipment, and construction

OPTICAL PROPERTIES

- Nanomaterials interact with light differently
- Color depends on particle size
- Enhanced light absorption and emission
- Key Idea:
  - Size changes color and light behavior
- Examples/Applications:
  - Quantum dots emit different colors depending on size
  - Used in LED screens, displays, and medical imaging

ELECTRONIC PROPERTIES

- Electron behavior changes at the nanoscale (quantum confinement)
- Conductivity can increase or change depending on material
- Materials may behave as conductors, semiconductors, or insulators
- Key Idea:
  - Electron movement is controlled at very small sizes
- Examples/Applications:
  - Microchips and transistors
  - Nanosensors and fast electronic devices

MAGNETIC PROPERTIES

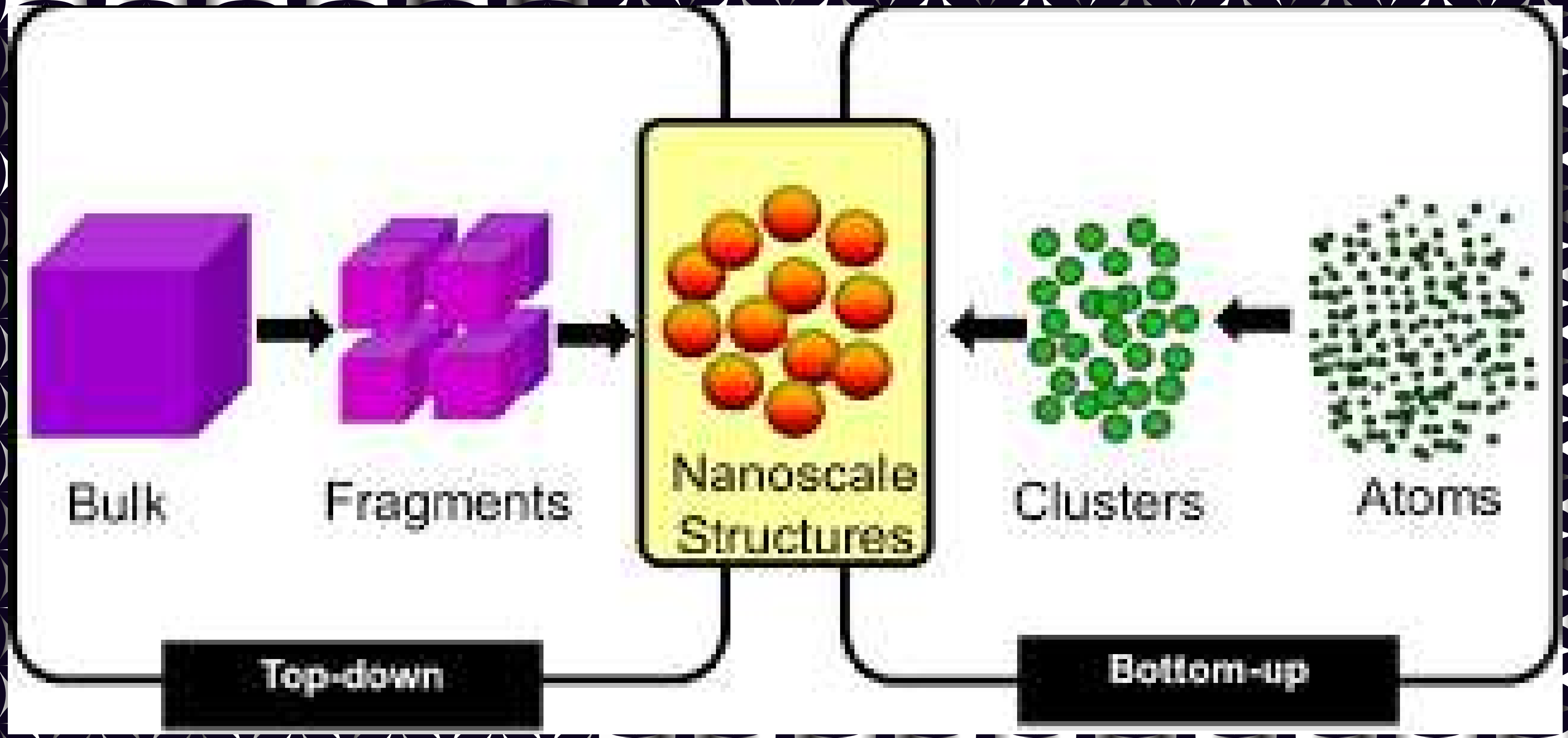
- Nanomaterials may show superparamagnetism
- Magnetic behavior changes at very small sizes
- Can be controlled using external magnetic fields
- Key Idea:
  - Magnetism becomes size-dependent
- Examples/Applications:
  - MRI contrast agents
  - High-density data storage

CHEMICAL PROPERTIES

- Nanomaterials are highly reactive due to large surface area
- More active sites for chemical reactions
- Faster reaction rates compared to bulk materials
- Key Idea:
  - Smaller size = faster and more efficient reactions
- Examples/Applications:
  - Catalysts in industrial processes
  - Pollution breakdown in environmental cleanup

Property	Key Characteristics at Nanoscale	Examples / Applications
Thermodynamic	Higher surface energy; lower melting point; less stable but more reactive	Gold nanoparticles with lower melting point; used in catalysts
Mechanical	Increased strength, hardness, and durability; often lightweight	Carbon nanotubes (stronger than steel); aerospace materials
Optical	Size-dependent color; strong light absorption and emission	Quantum dots in displays and medical imaging
Electronic	Altered electron behavior (quantum effects); variable conductivity	Microchips, transistors, nanosensors
Magnetic	Superparamagnetism; magnetic behavior depends on size	MRI contrast agents; data storage devices
Chemical	Very high reactivity due to large surface area and active sites	Catalysts; pollution control and environmental cleanup

SYNTHESIS OF NANO MATERIALS

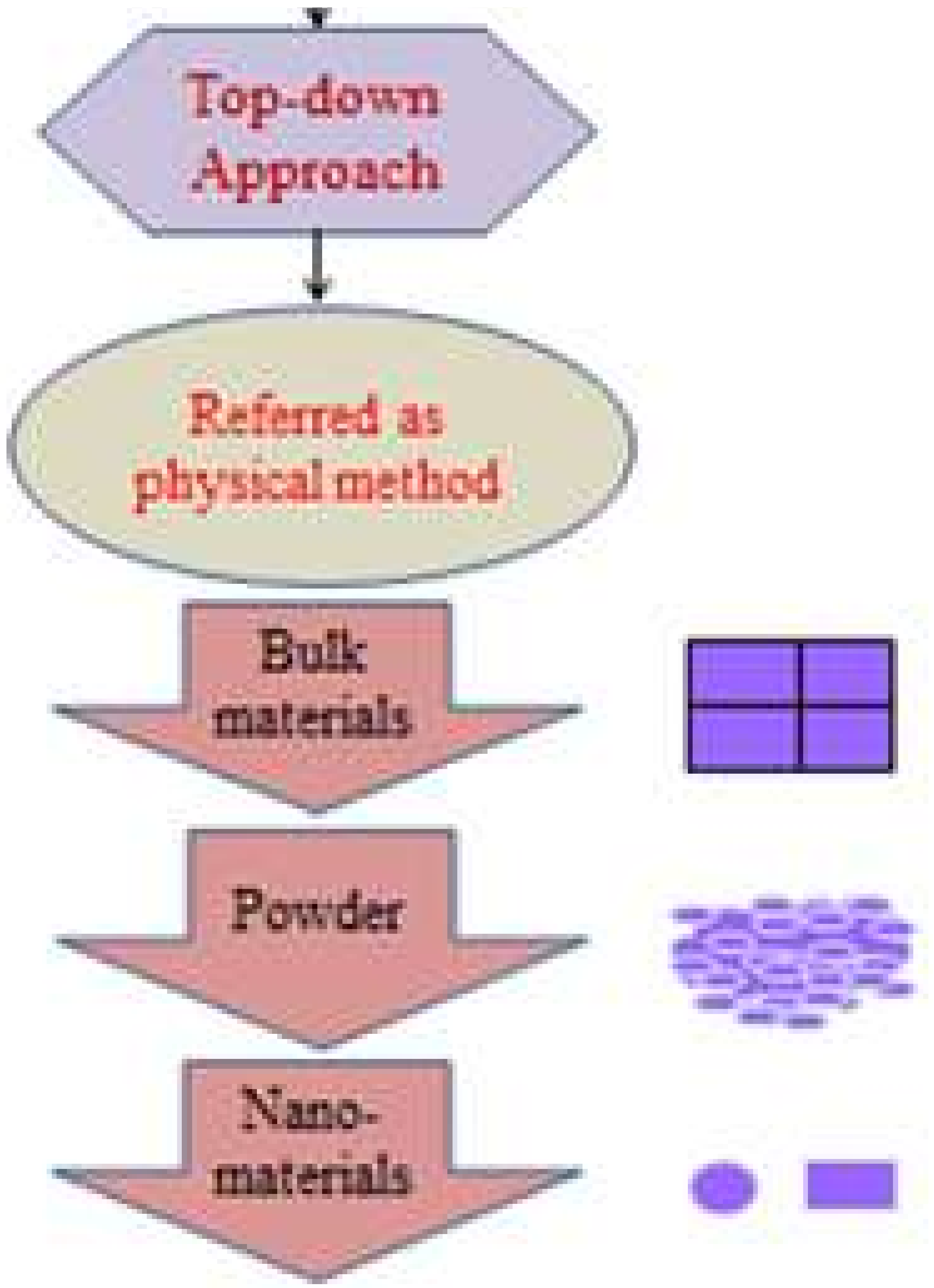


TOP DOWN APPROACH:

Subtractive Synthesis

APPROACHES

TOP DOWN APPROACH

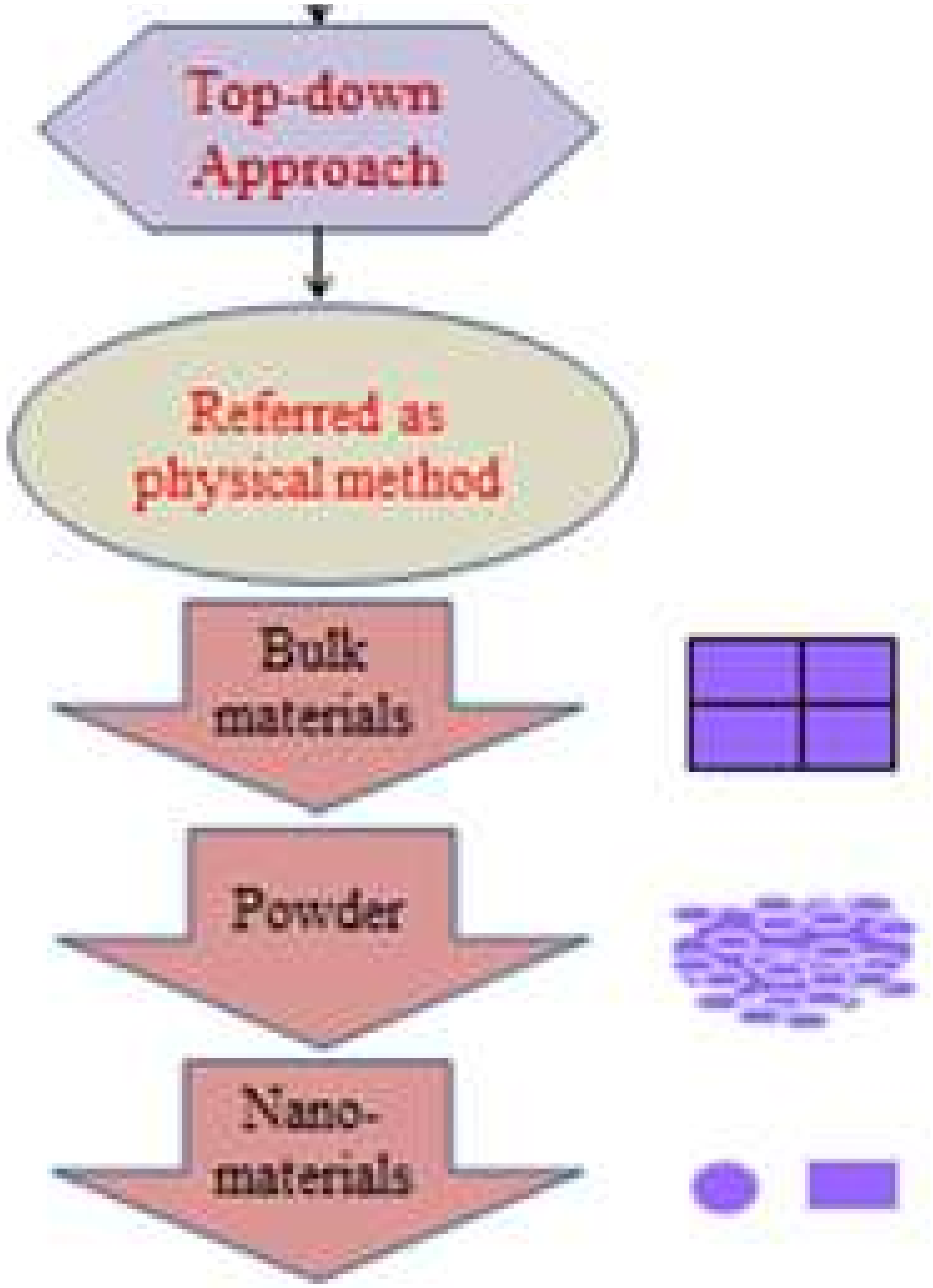


A ***subtractive method*** where *bulk materials* are **physically** or *chemically* broken down into nano-sized particles
(1-100nm)

-superstructures To nano particles

APPROACHES

TOP DOWN APPROACH



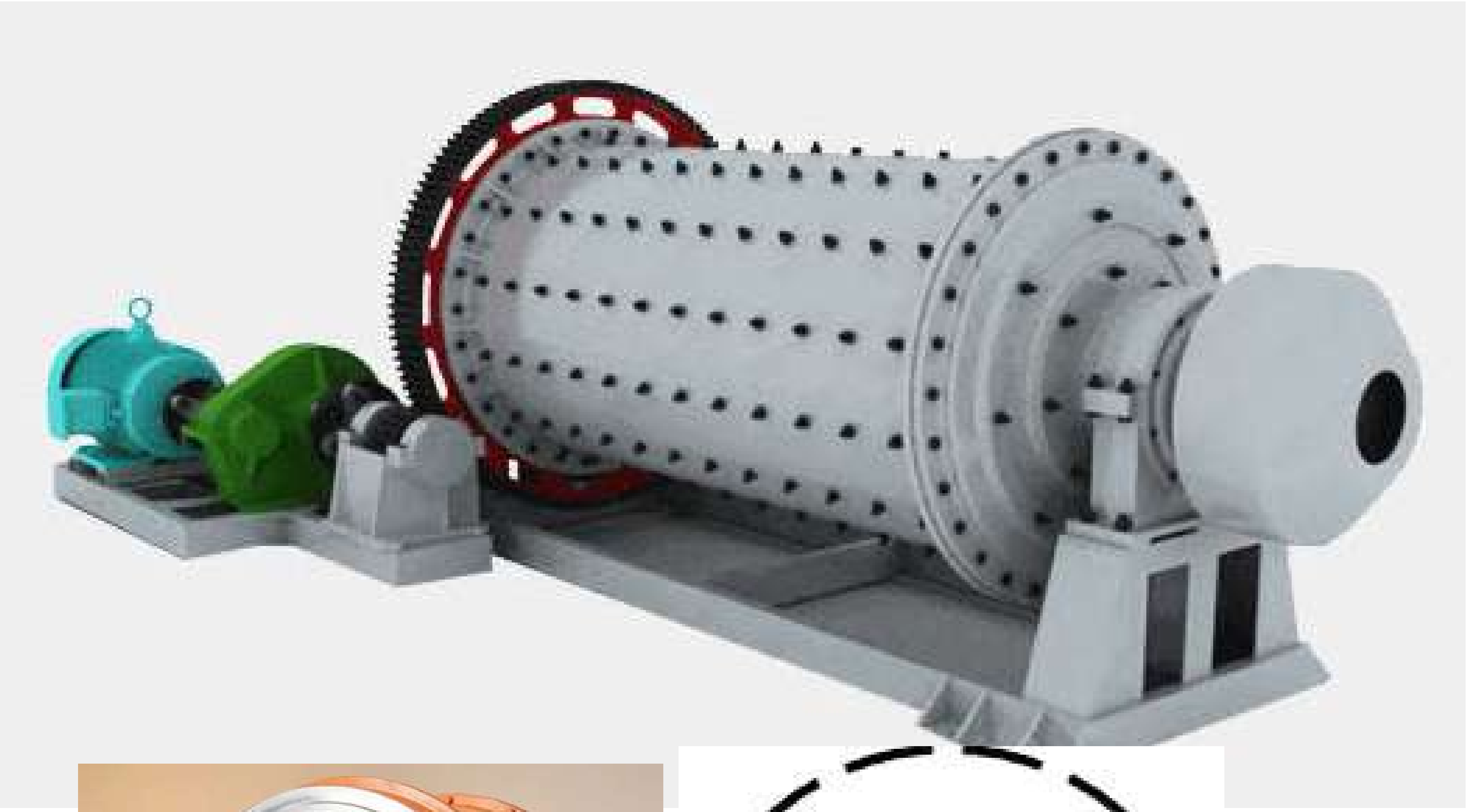
General limitations:

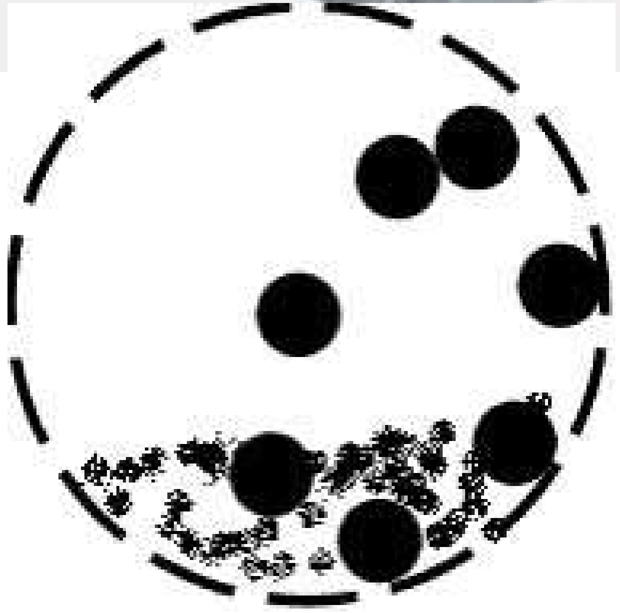
Difficulty in maintaining precise size and shape uniformity.

- Tendency to produce ***irregular particle morphologies.***

TOP DOWN APPROACH METHODS

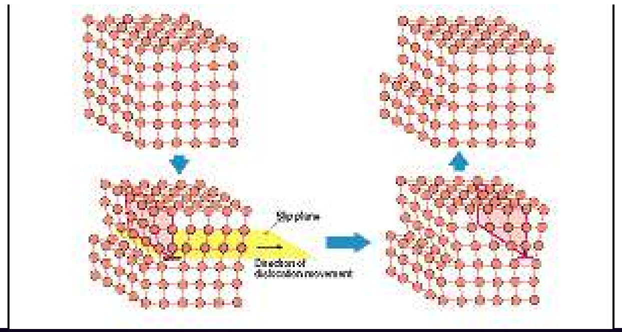
Mechanical milling or ball milling process





Kinetic energy is transferred from grinding media (balls) to the precursor material (bulk media) within a vessel.

High *pressure and temperature* lead to **phase transformation**, crystal size reduction, and mechanical dislocation.



TOP DOWN APPROACH METHODS

Mechanical milling or ball milling process

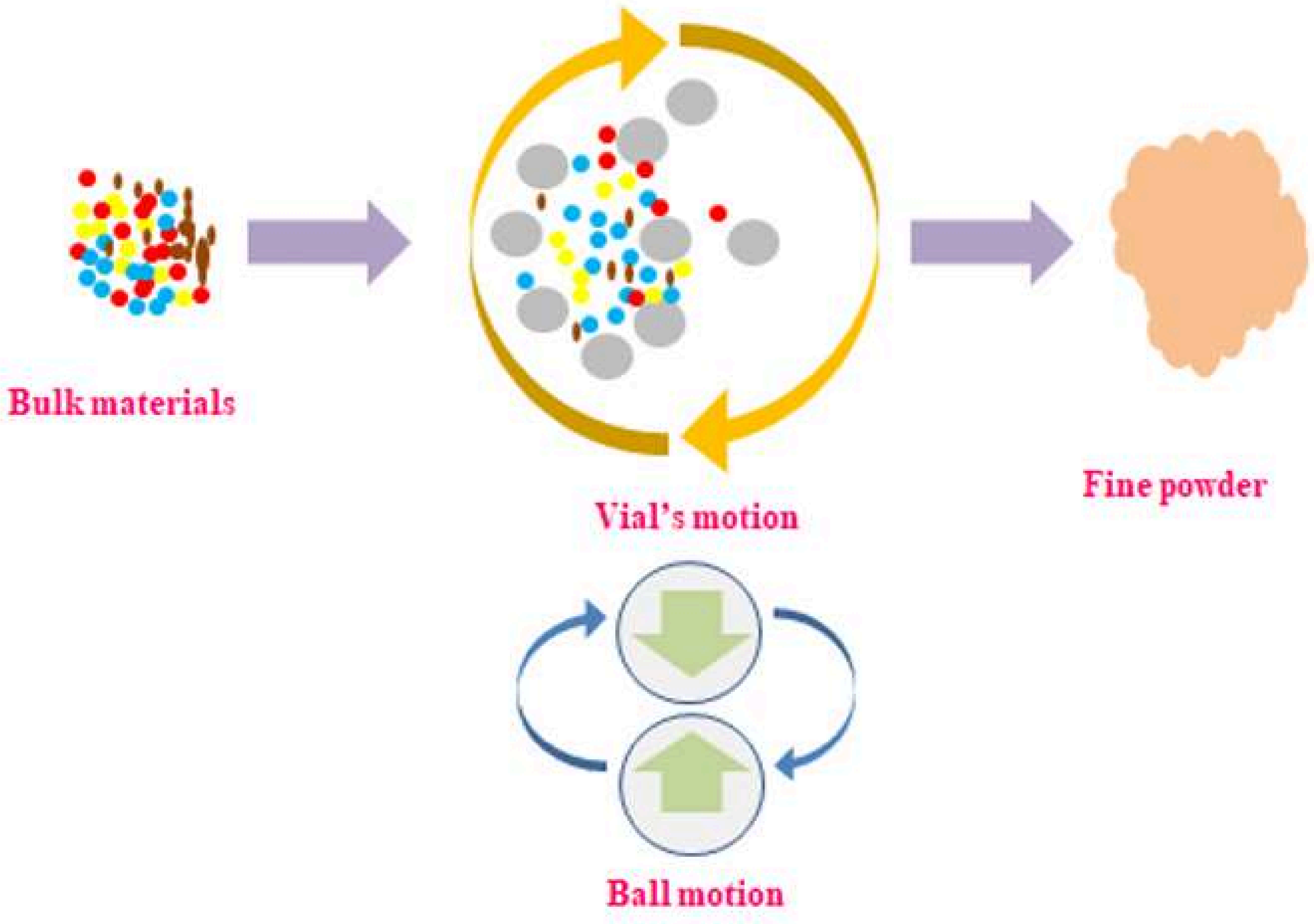


Figure 4. Schematics of ball milling process

Pros: Universal application, high capacity, reliable, and cost-effective for long-term maintenance.

Cons: Significant energy wastage, risk of contamination (from the milling balls/vessel), and potential to disorder the crystal structure.

APPROACHES

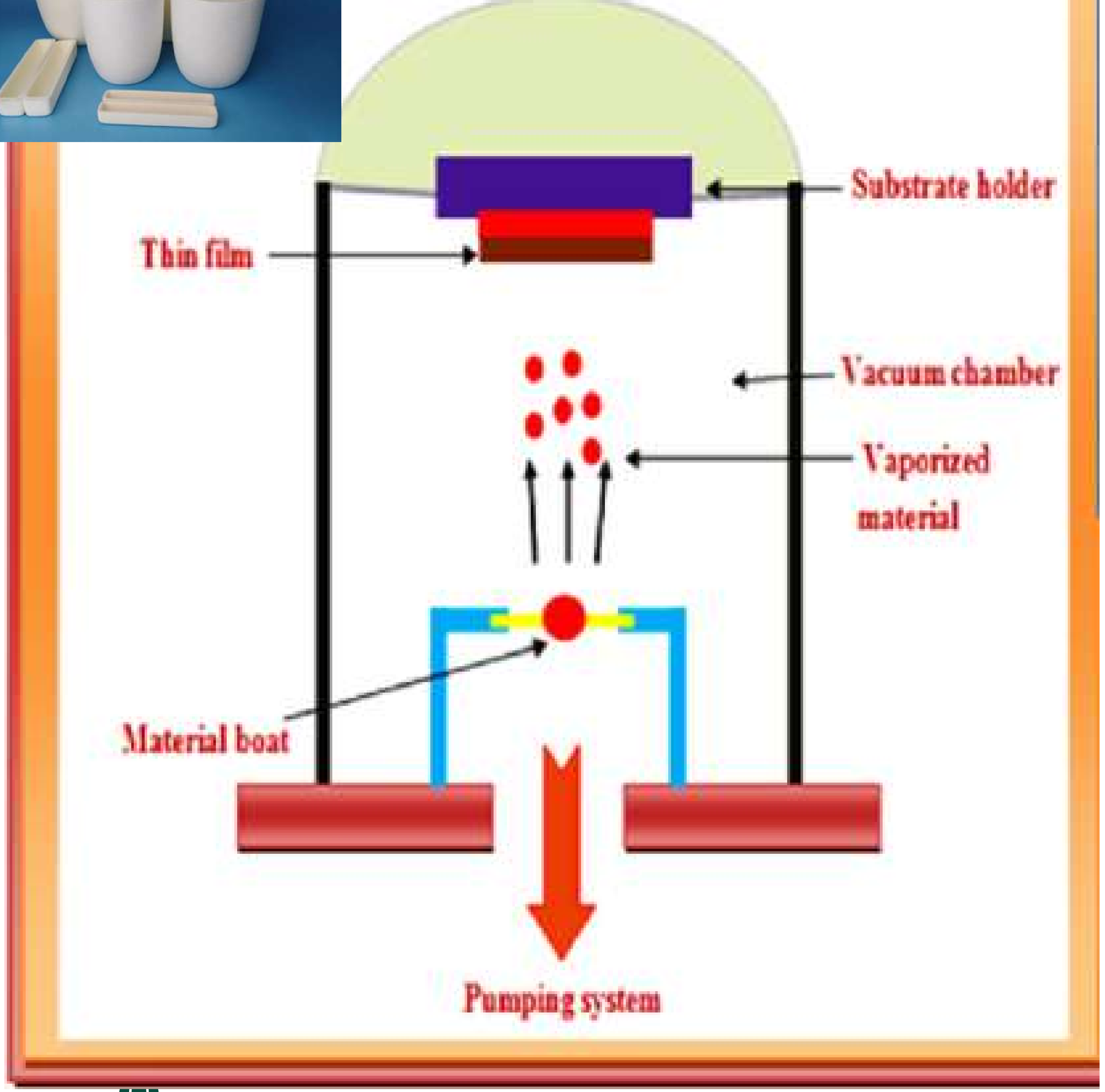
TOP DOWN APPROACH METHODS

Thermal Evaporation

An ***endothermic process*** where heat breaks molecular bonds to create a vapor, which then condenses into nanoparticles.

APPROACHES

TOP DOWN APPROACH METHODS



Thermal Evaporation

-Utilizes r***esistive heating in an alumina crucible.***

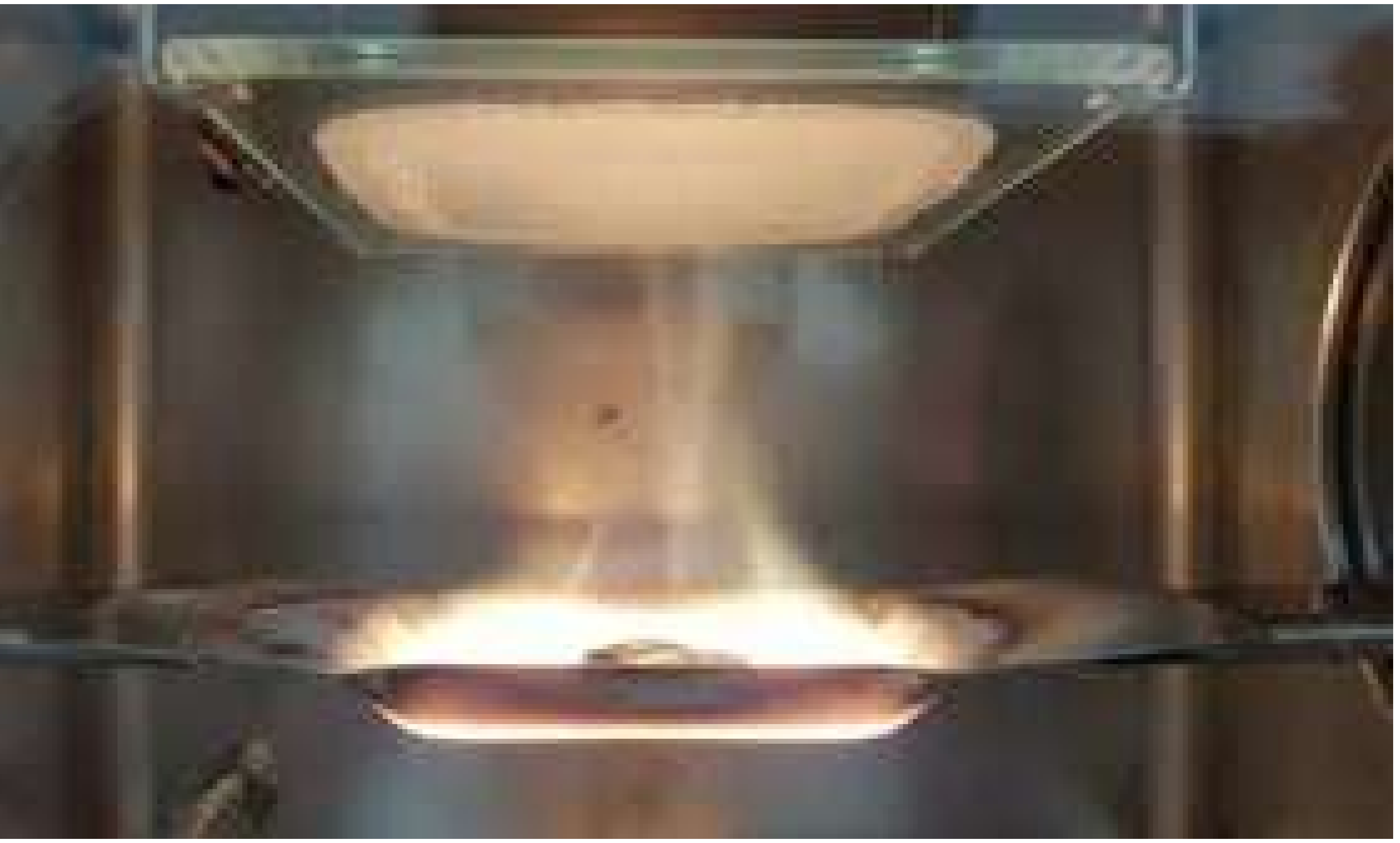
Controlled by **PID (Proportional Integral Derivative)** controllers for precise temperature regulation.

Substrates must be **ultrasonically cleaned** (acetone/ethanol) to ensure deposition quality.

APPROACHES

TOP DOWN APPROACH METHODS

Thermal Evaporation

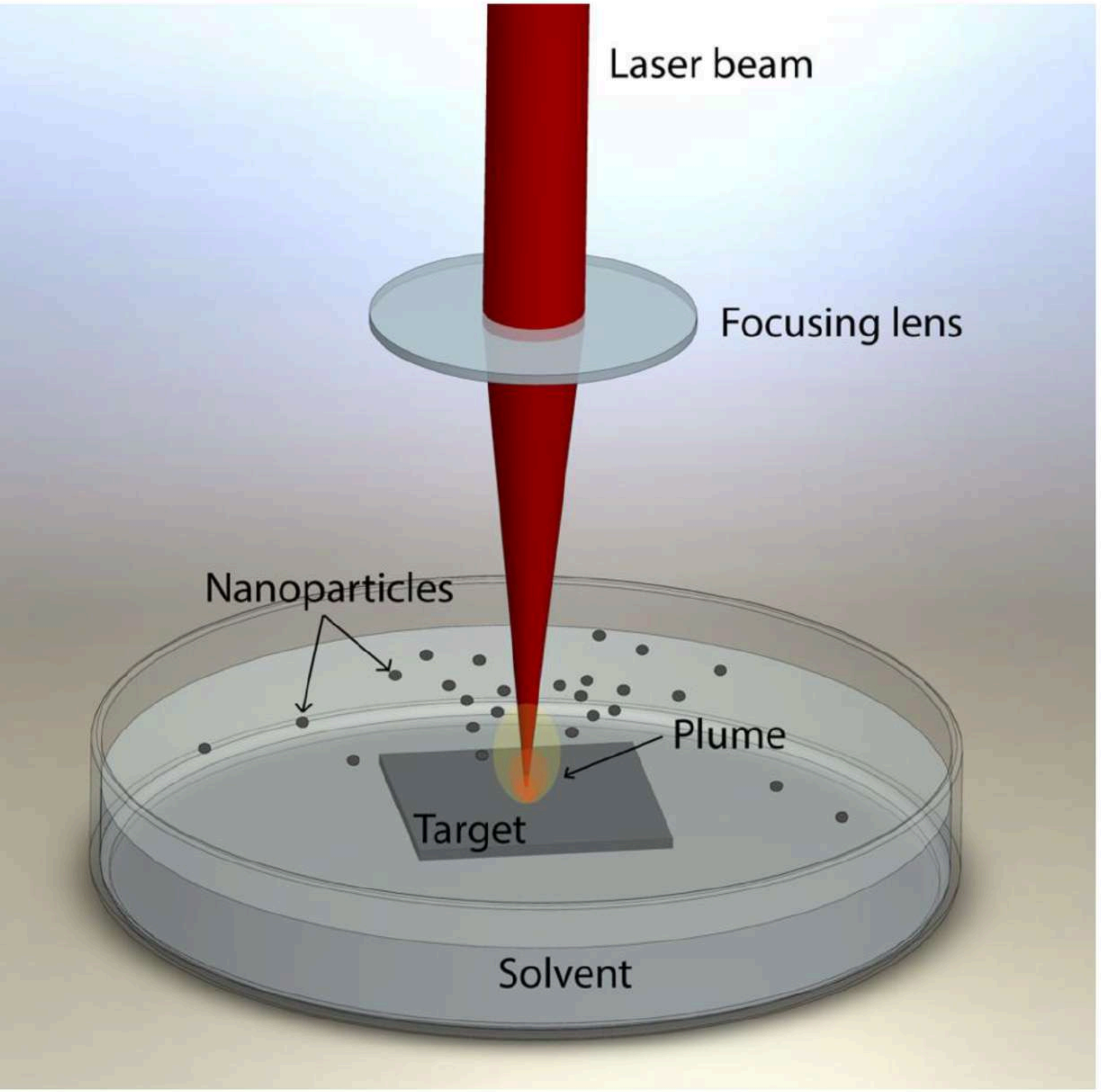


- Pros: Produces monodispersed (uniform size) suspensions; “solvent-free“ mode prevents chemical impurities.
- Cons: Difficult to deposit *complex alloys;* quality is highly dependent on the purity of the source material.

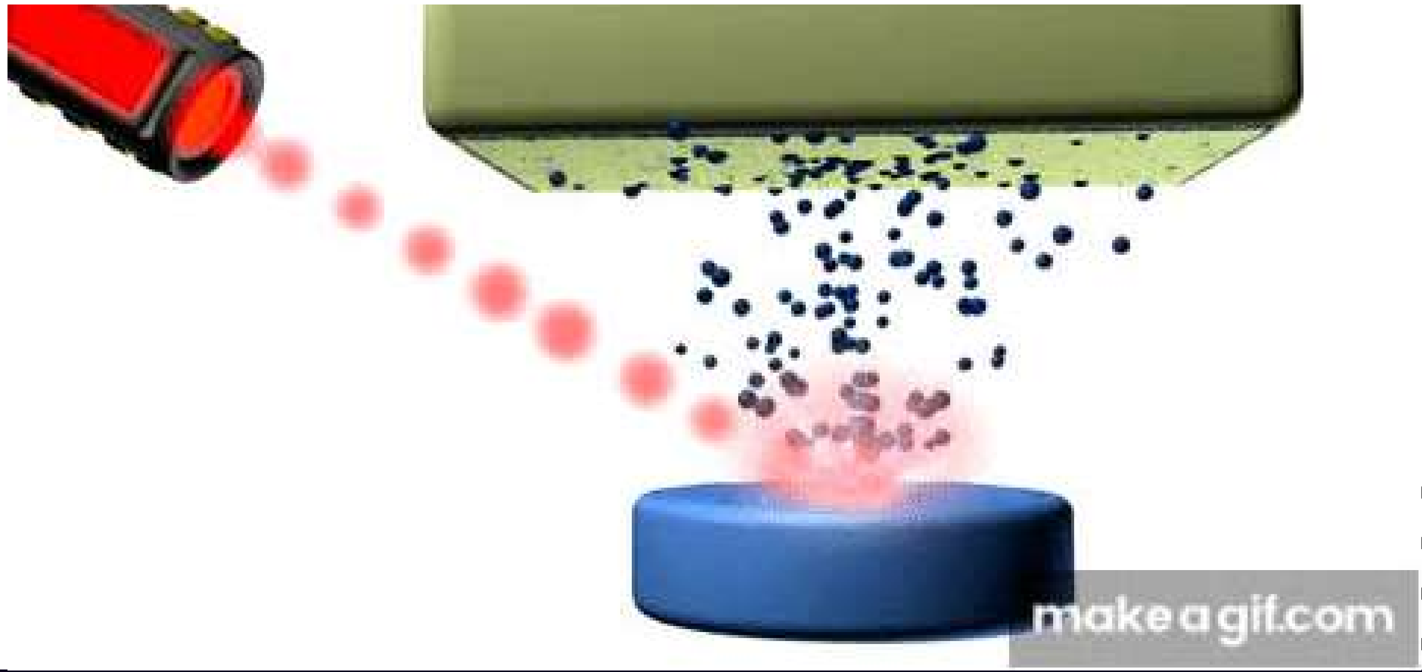
APPROACHES

TOP DOWN APPROACH METHODS

Laser Ablation



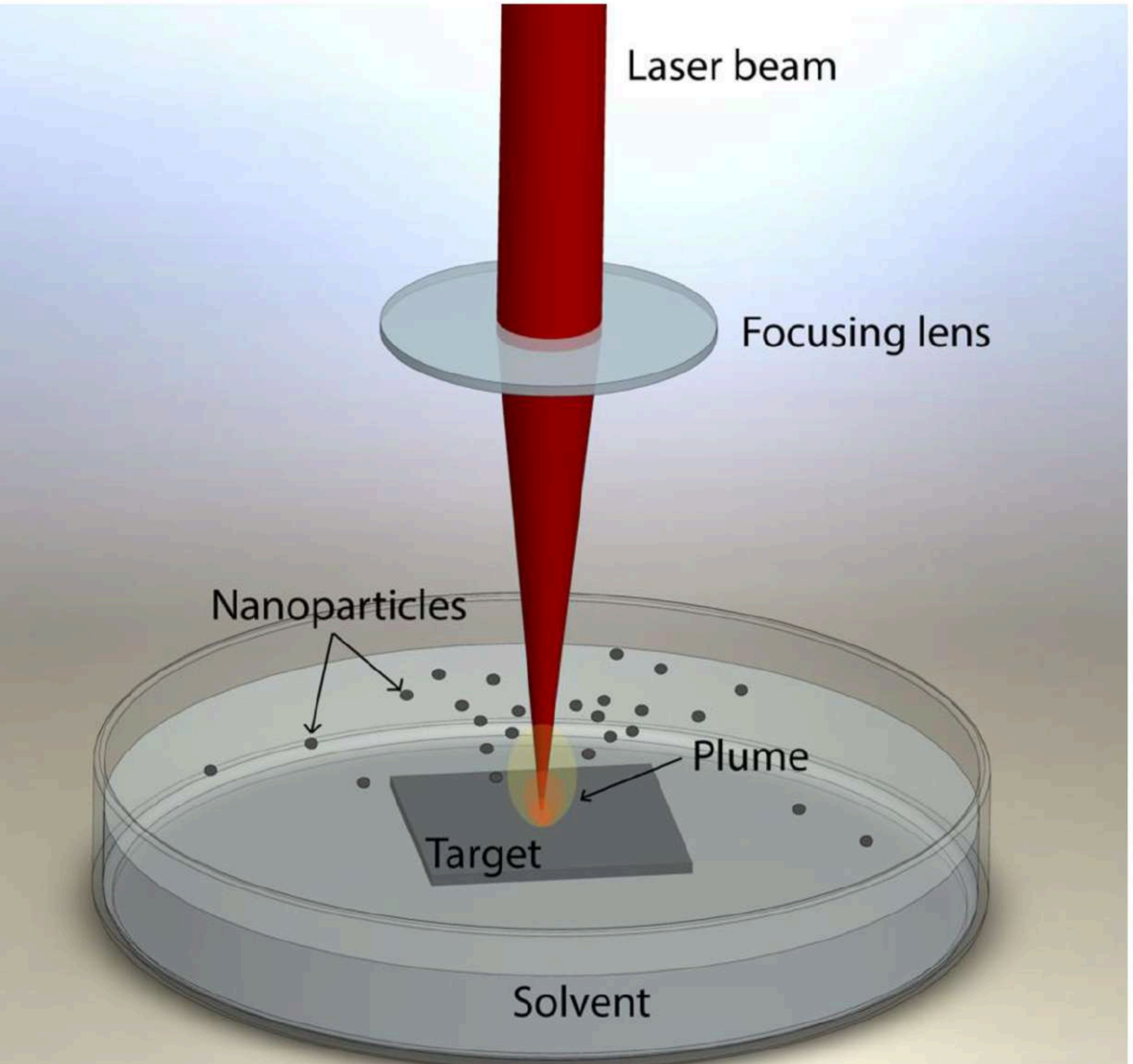
A **pulsed electromagnetic beam** (laser) is focused on a target (often submerged in liquid). The material absorbs energy, leading to **melt ejection, evaporation, and vaporization.**



APPROACHES

TOP DOWN APPROACH METHODS

Laser Ablation



Deionized water or ethanoll

Key advantage: ligand-free (pure surface) noble metal particles.

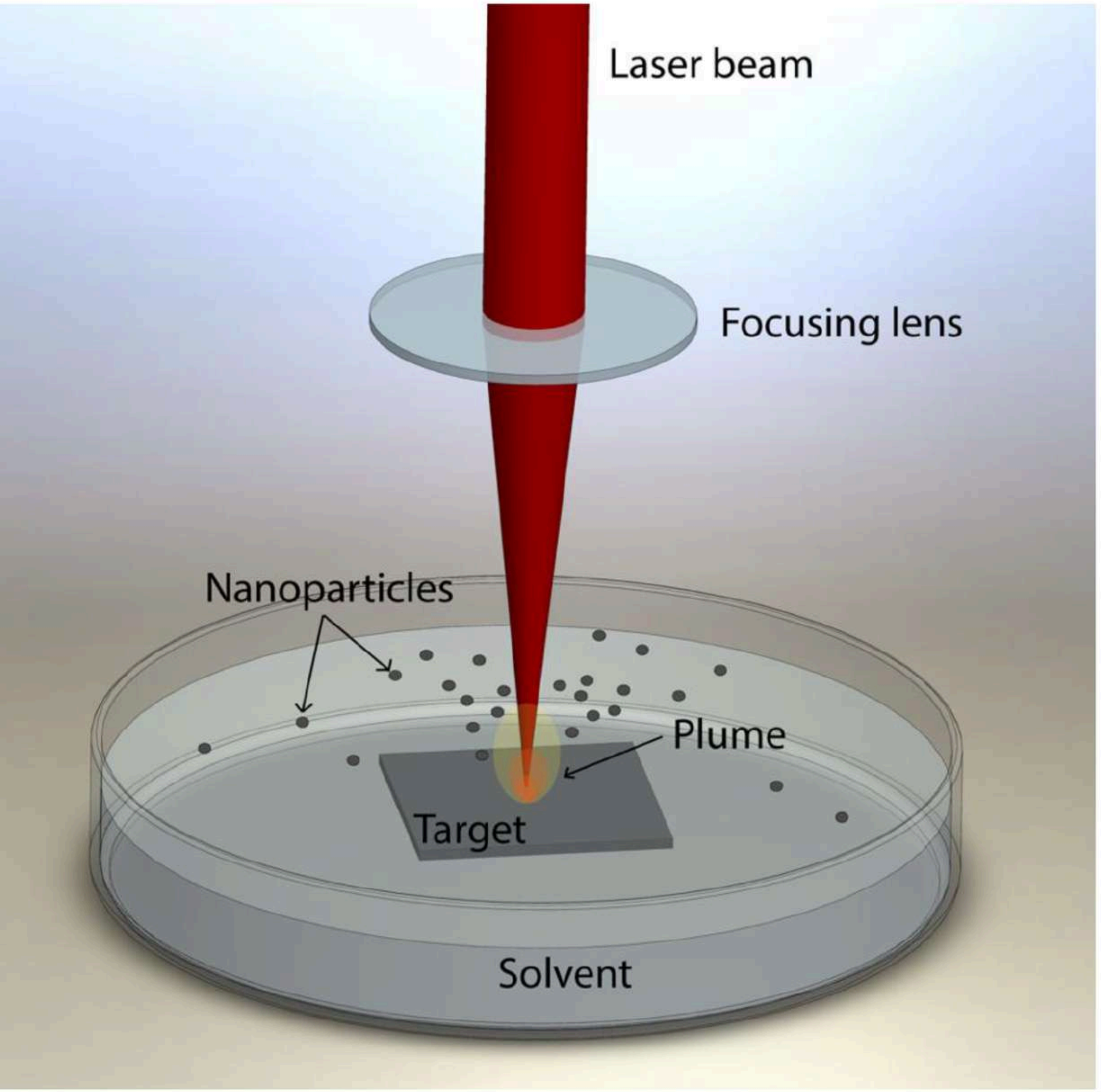
Adjustability: Particle size and morphology are tuned by modifying laser settings (fluence/pulse) and the liquid medium.

agglomerate is a cluster of nanoparticles or smaller aggregates held together by relatively weak physical forces, such as van der Waals forces, electrostatic interactions, or capillary forces

APPROACHES

TOP DOWN APPROACH METHODS

Laser Ablation

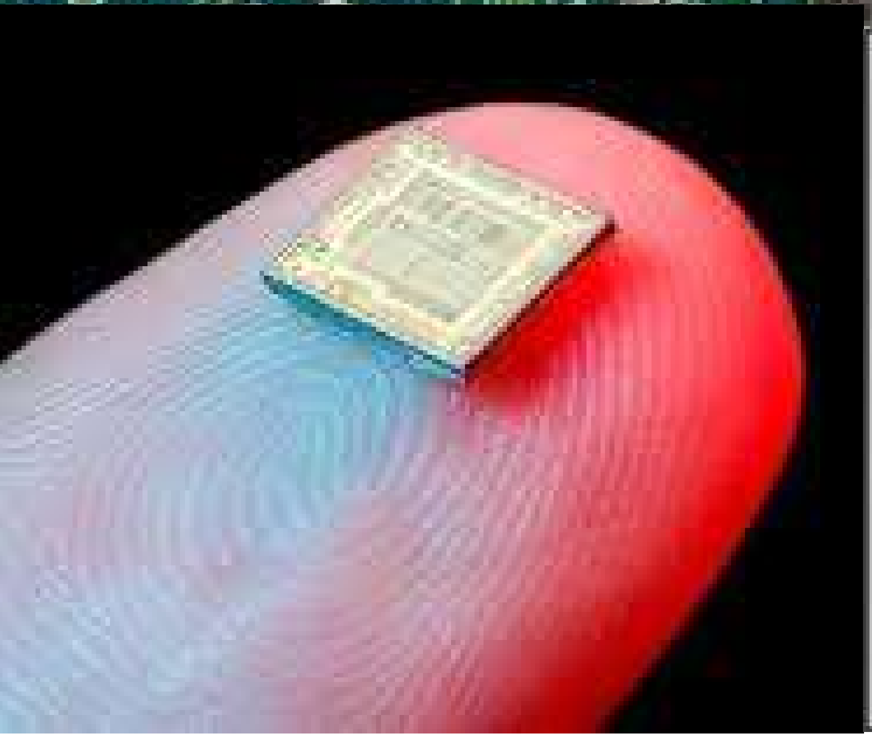


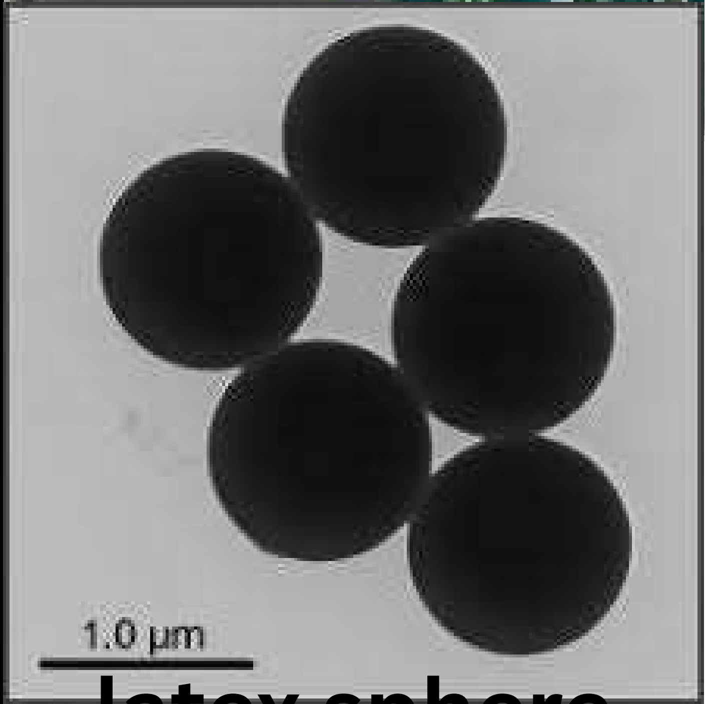
·Pros: Fast processing speed and low energy loss during the ablation itself.

·Cons: High initial energy requirement; *scalability is difficult* because the laser source disperses over industrial-scale volumes.

APPROACHES

TOP DOWN APPROACH METHODS





latex sphere

Lithographic Methods

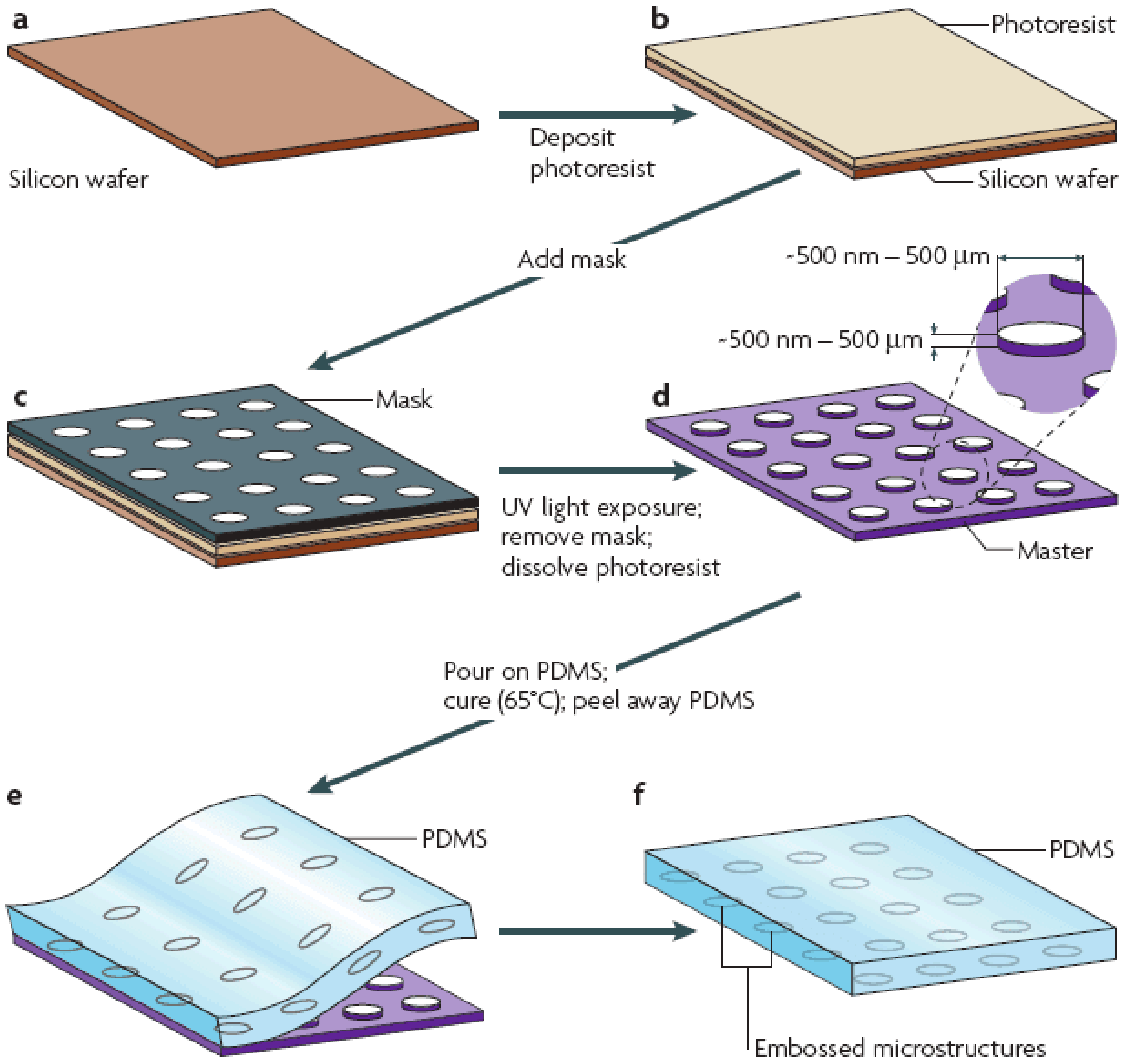
A patterning technique originally used for **printed circuits**, now **adapted for nano-fabrication.**

Nanoimprint Lithography: Moves away from light-based etching to a **stamping method**—creating a template matric (e.g., latex spheres) and stamping soft polymers to form patterns.

APPROACHES

TOP DOWN APPROACH METHODS

Lithographic Methods



- Electron Beam (E-beam) & Focused Ion Lithography.
- Dip-pen & **Soft Lithography.**

Schematic illustration of Soft Lithography and Pattern Transfer using PDMS

APPROACHES

TOP DOWN APPROACH METHODS

Lithographic Methods

- Pros: Capable of creating highly structured, precise micron-to-nano sized patterns.

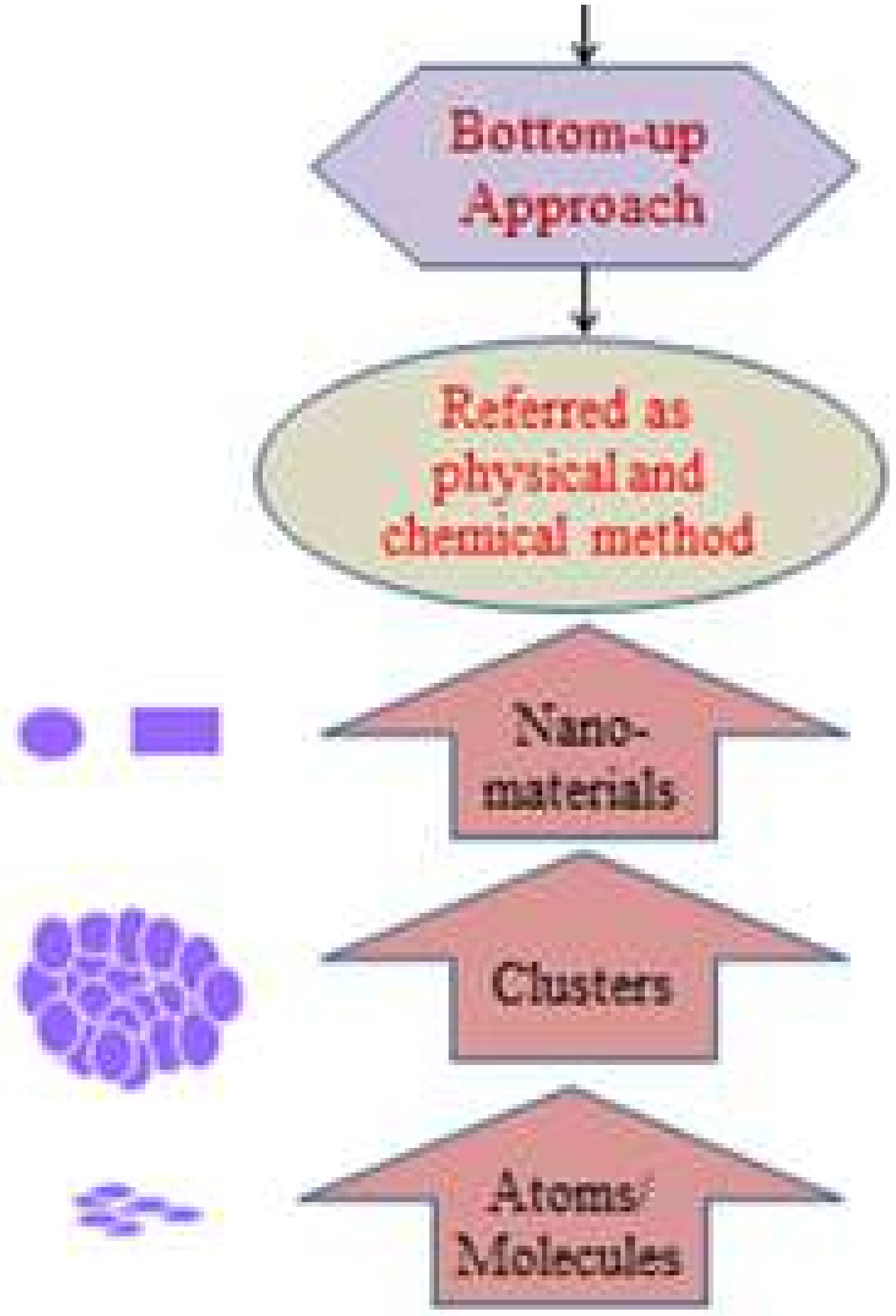
- Cons: Requires prohibitively expensive equipment and has high energy demands.

BOTTOM UP APPROACH:

Constructive Synthesis

APPROACHES

BOTTOM UP APPRACH



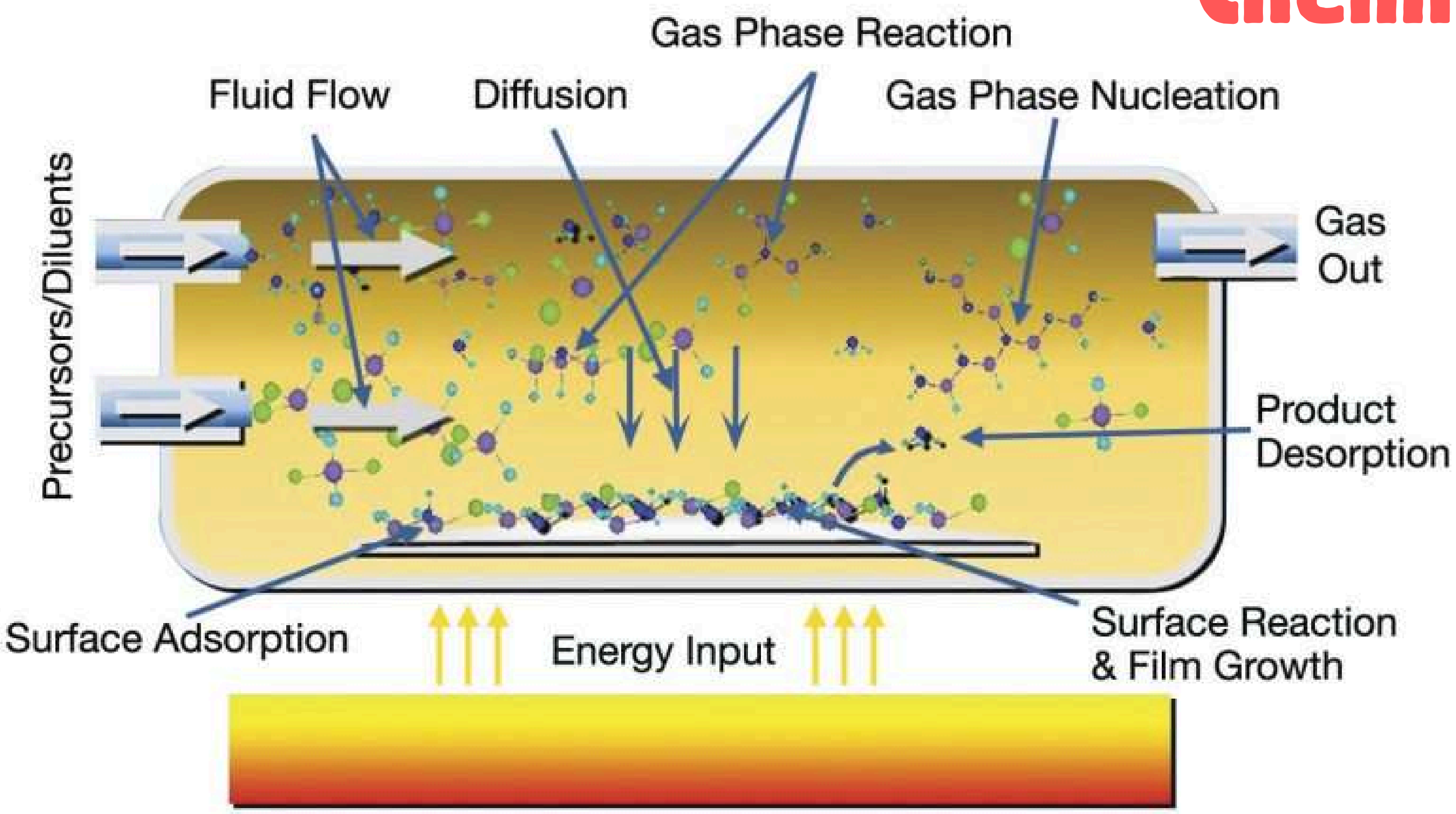
An **additive method** where atoms or molecules act as “building blocks“ to assemble into complex nanostructures.

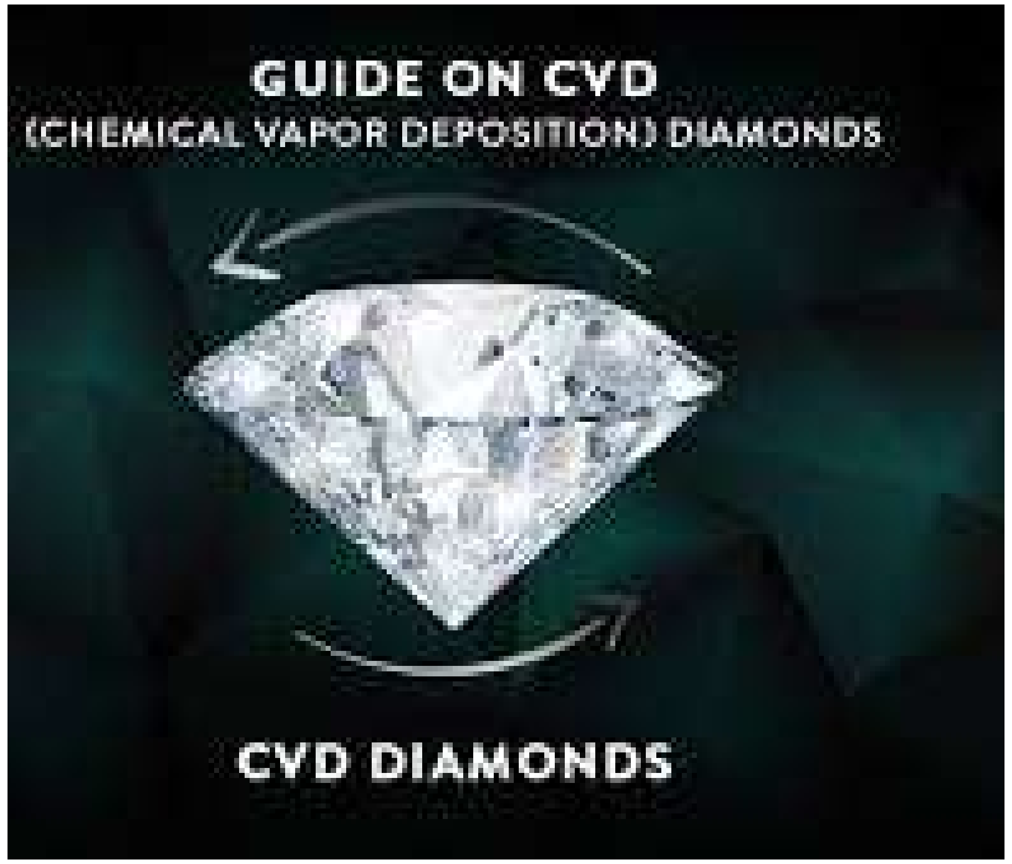
Relies on ***chemical reactions and self-assembly*** of building units.

APPROACHES

BOTTOM UP APPRACH

Chemical Vapor Deposition (CVD)





lab grown diamonds form Methane CH4 gas

- Gaseous reactants are introduced into a reaction chamber. When they contact a heated substrate, a chemical reaction occurs, depositing a thin solid film.

APPROACHES

BOTTOM UP APPRACH

Chemical Vapor Deposition (CVD)

Critical Parameters:

- Precise delivery of gas-phase reactants.
- Monitoring of reaction pressure and energy source.
- Strict automation and cleaning of exhaust gases (safety protocol).

- Pros: Produces exceptionally pure, dense, and robust materials; repeatable growth rate; fine crystal quality with minimal flaws.
- Cons: Precursors are often poisonous, flammable, or expensive; high deposition temperatures limit the types of substrates that can be used.

APPROACHES

BOTTOM UP APPRACH

Hydrothermal Synthesis

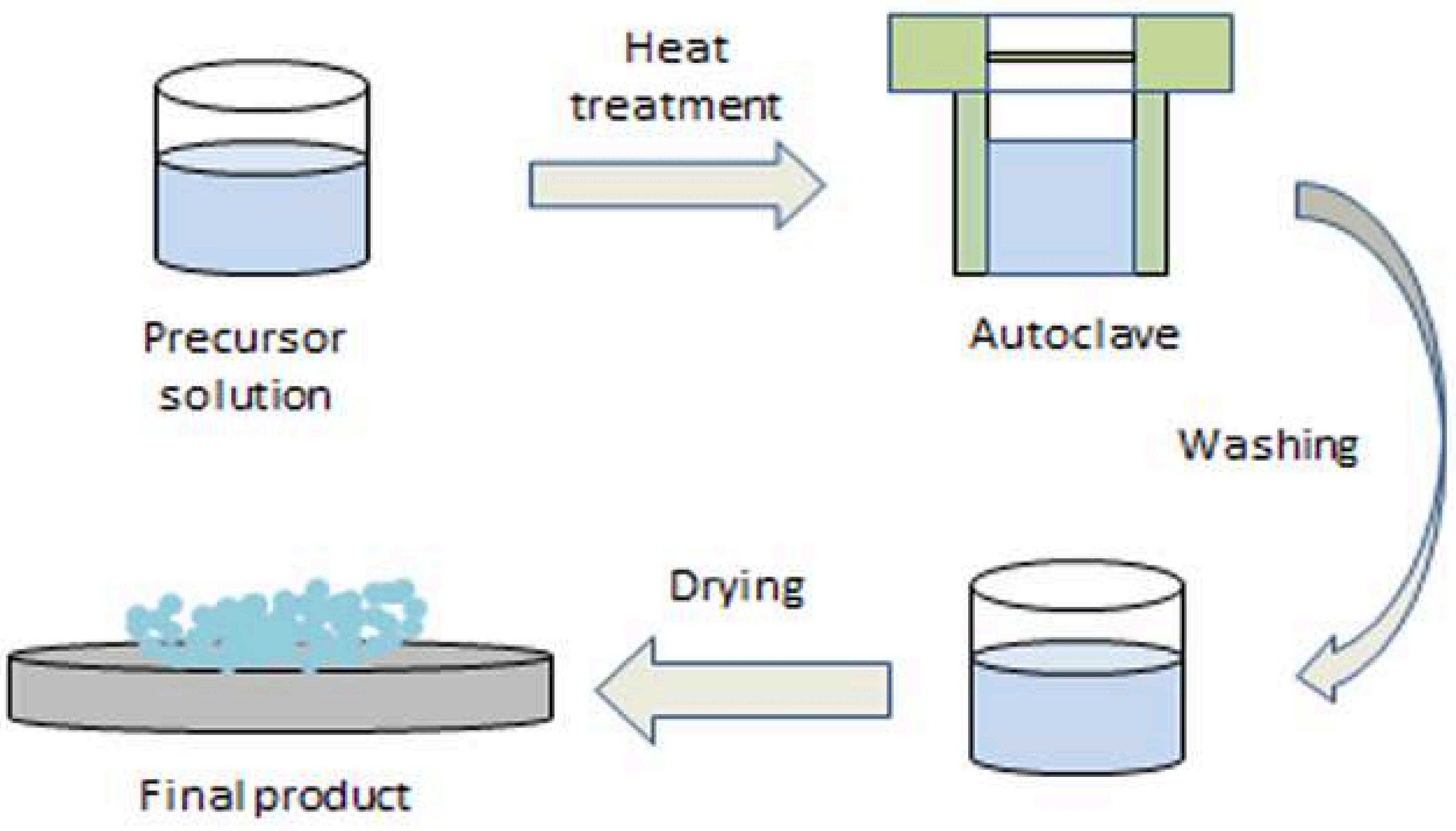


Figure 6. Schematics of hydrothermal technique

A solution-based approach using **water as a solvent** in a high-pressure steel vessel called an **autoclave (Teflon-lined stainless steel autoclave)**

- The temperature is ***raised above water's boiling point (supercritical conditions)*** to reach vapor saturation.

APPROACHES

BOTTOM UP APPRACH

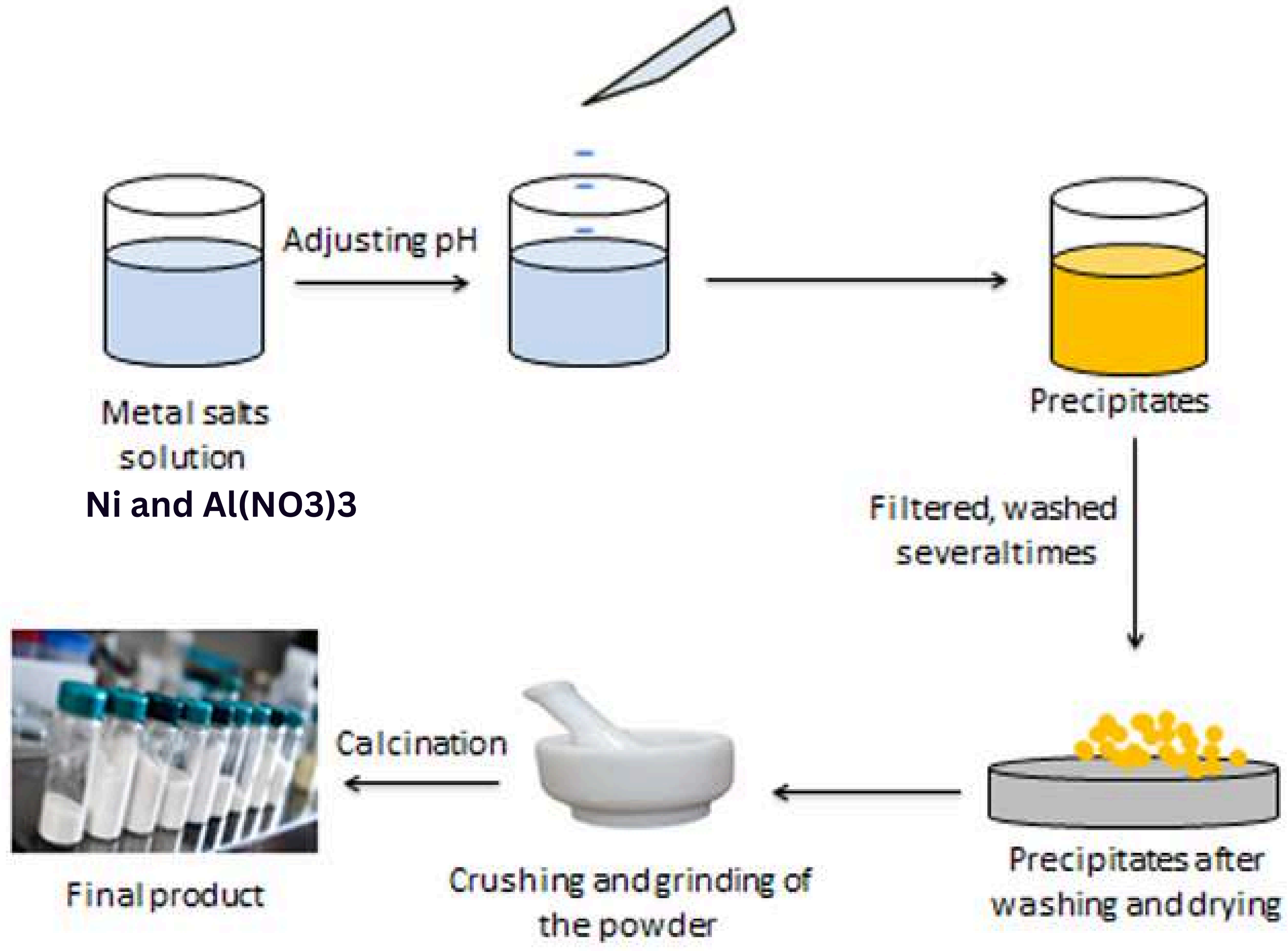
Hydrothermal Synthesis

Unique Output:
Capable of synthesizing specialized materials like **luminescent phosphorus, super-ionic conductors, and microporous crystals.**

APPROACHES

BOTTOM UP APPRACH

Co-Precipitation Method



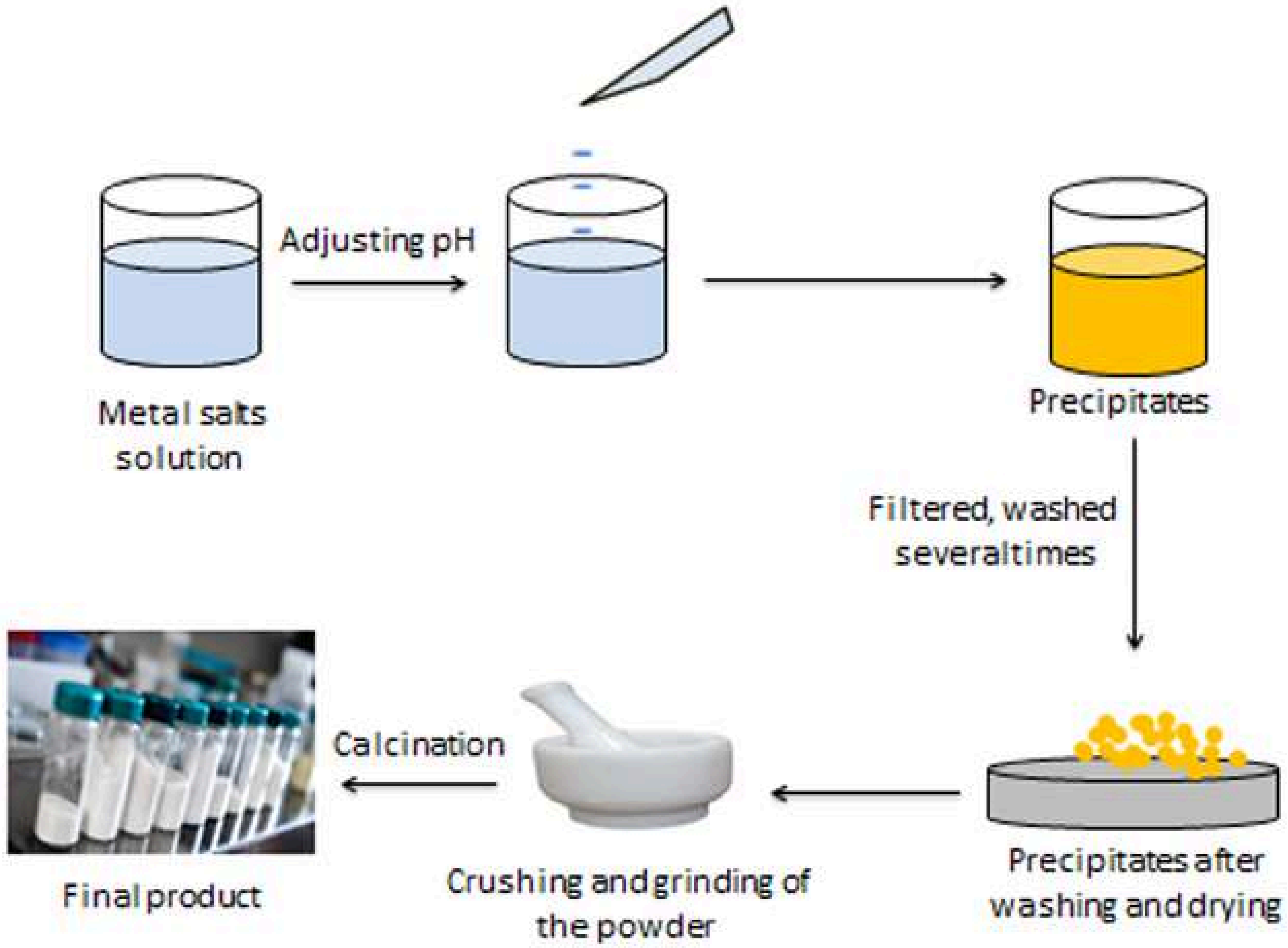
- A “**wet chemical**“ process where two or more cations in a solution are precipitated together.

- One solution is added drop-by-drop into another containing precipitating agents (e.g., *hydroxides, chlorides, or oxalates*).
- Post-Processing: **Requires rigorous filtration and washing** to separate the nanomaterial from impurities.

APPROACHES

BOTTOM UP APPRACH

Co-Precipitation Method

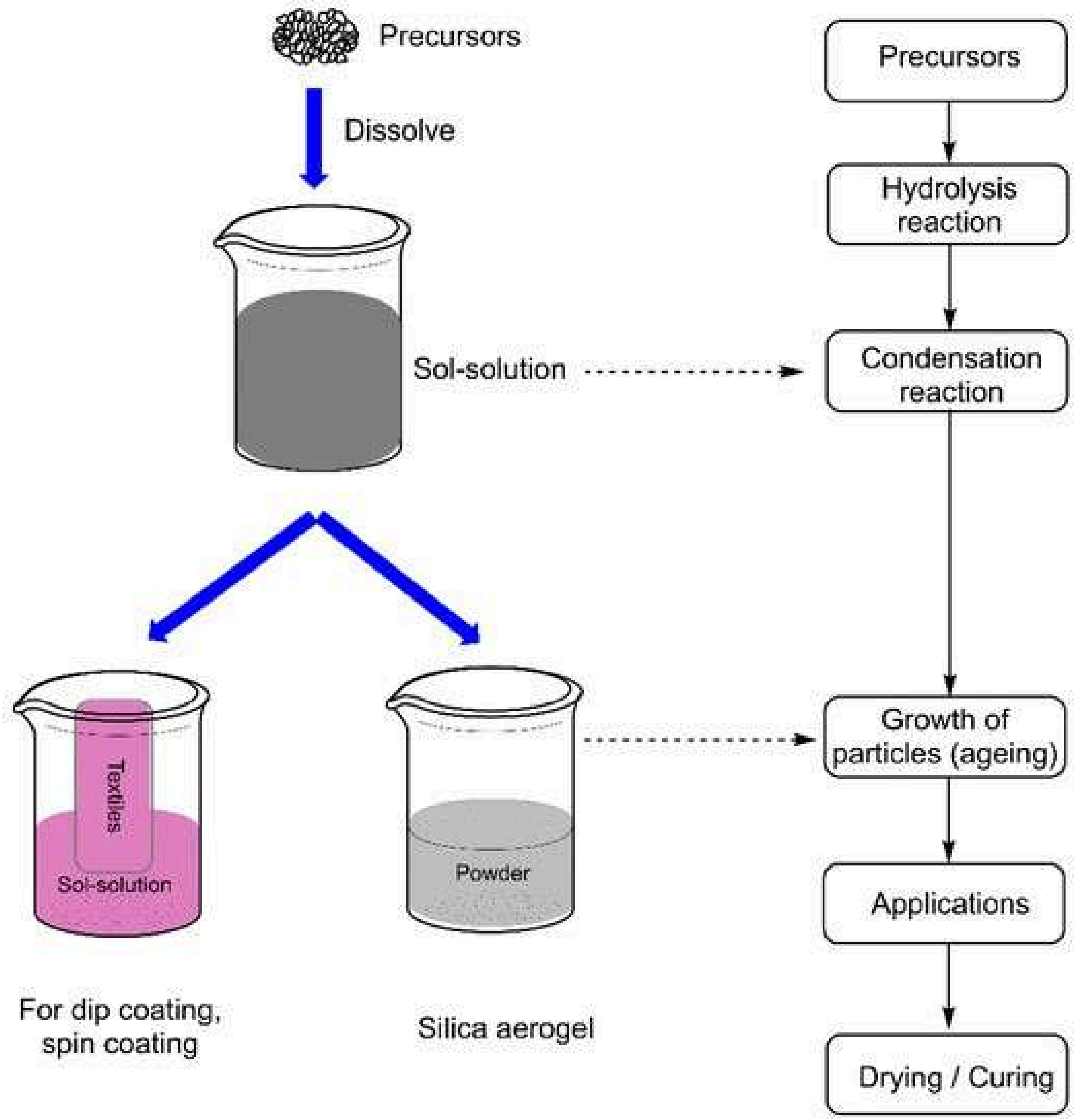


- Pros: The simplest and quickest method; energy-efficient and maintains homogeneity across the sample.
- Cons: Difficult to manage different precipitation rates between components; uses hazardous chemicals that pose environmental and health risks

APPROACHES

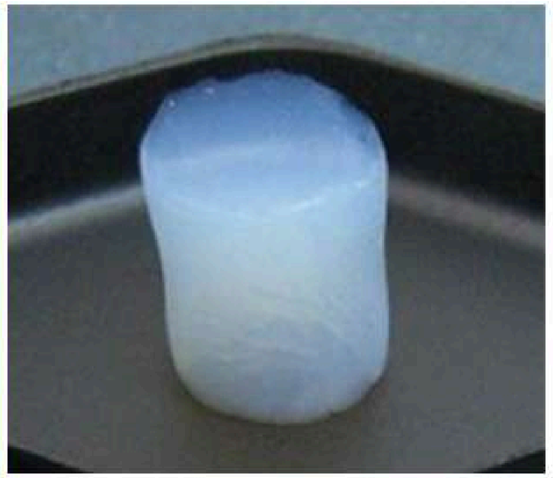
BOTTOM UP APPRACH

Sol-Gel Method



- A versatile "Wet Chemical" technique involving the **transition of a system from a Sol** (a *colloidal suspension* of solid particles) to a **Gel** (an integrated solid macromolecular network within a liquid phase).

Porous glass from classical sol-gel



Glass foam (aerogel)

Hybrid glass (organic-inorganic) made by fast sol-gel process

APPROACHES

BOTTOM UP APPRACH

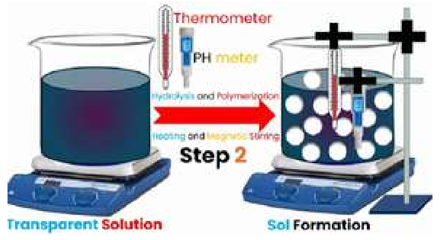
Sol-Gel Method (5 step process)

1. **Hydrolysis**: The precursor (typically metal alkoxides) reacts with water or alcohol to form a hydroxide solution.

Hydrolysis reaction occurs result in colloidal solution (sol)

$$M(OR)_n + H_2O \rightarrow M(OH)_n + R(OH)$$

2. **Polycondensation**: Removal of water/alcohol facilitates the formation of metal-oxide bonds, increasing viscosity until a gel state is achieved.



3. **Aging:** Continuous condensation within the gel network strengthens the structure and reduces porosity over time.

4. **Drying:** A critical phase where water and organic components are removed; careful handling is required to preserve the structural integrity.

5. **Calcination:**

A final thermal treatment used to eliminate residues and obtain the pure, crystalline final product. (300-800 degree Celsius)

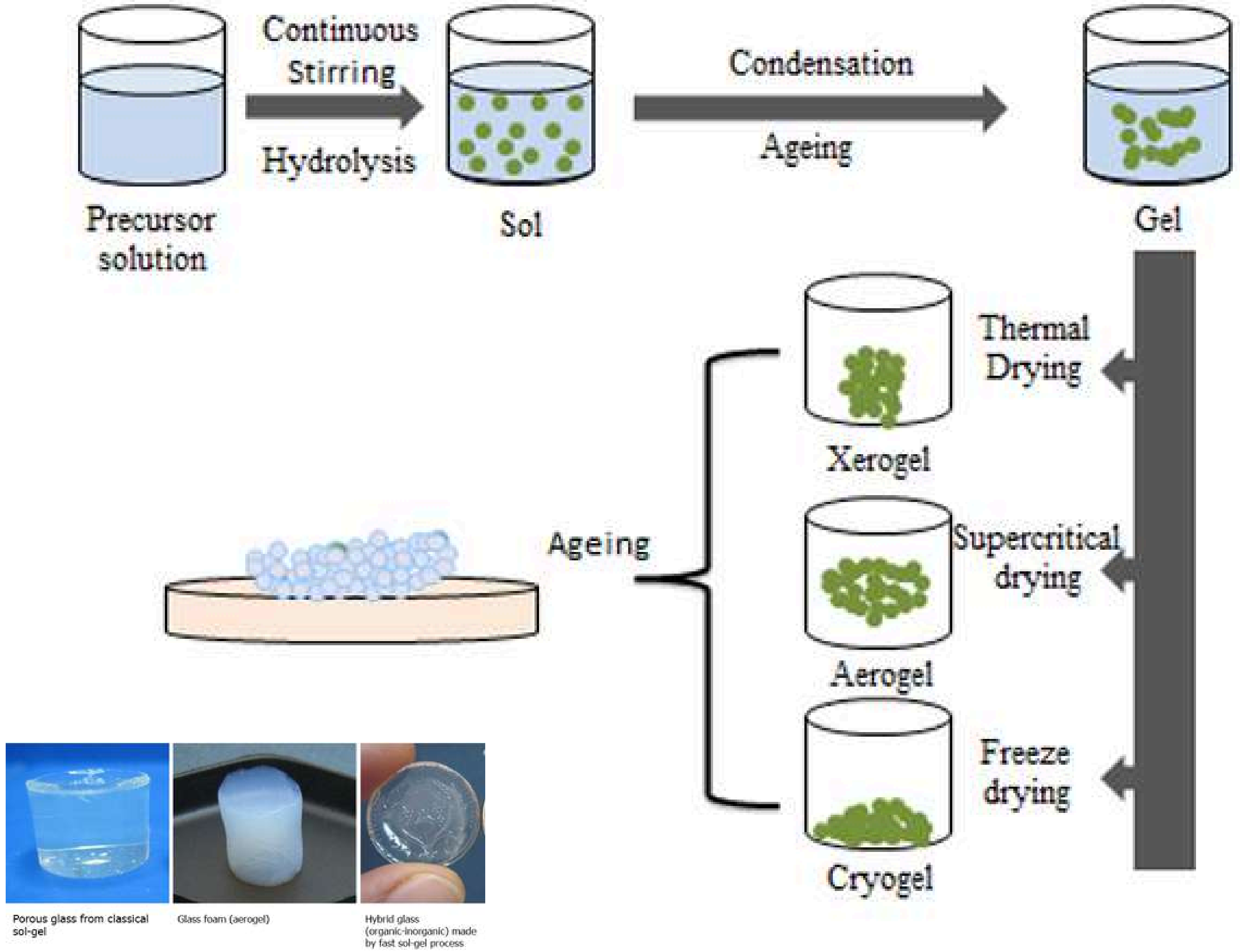


Figure 8. Schematics of sol-gel technique

APPROACHES

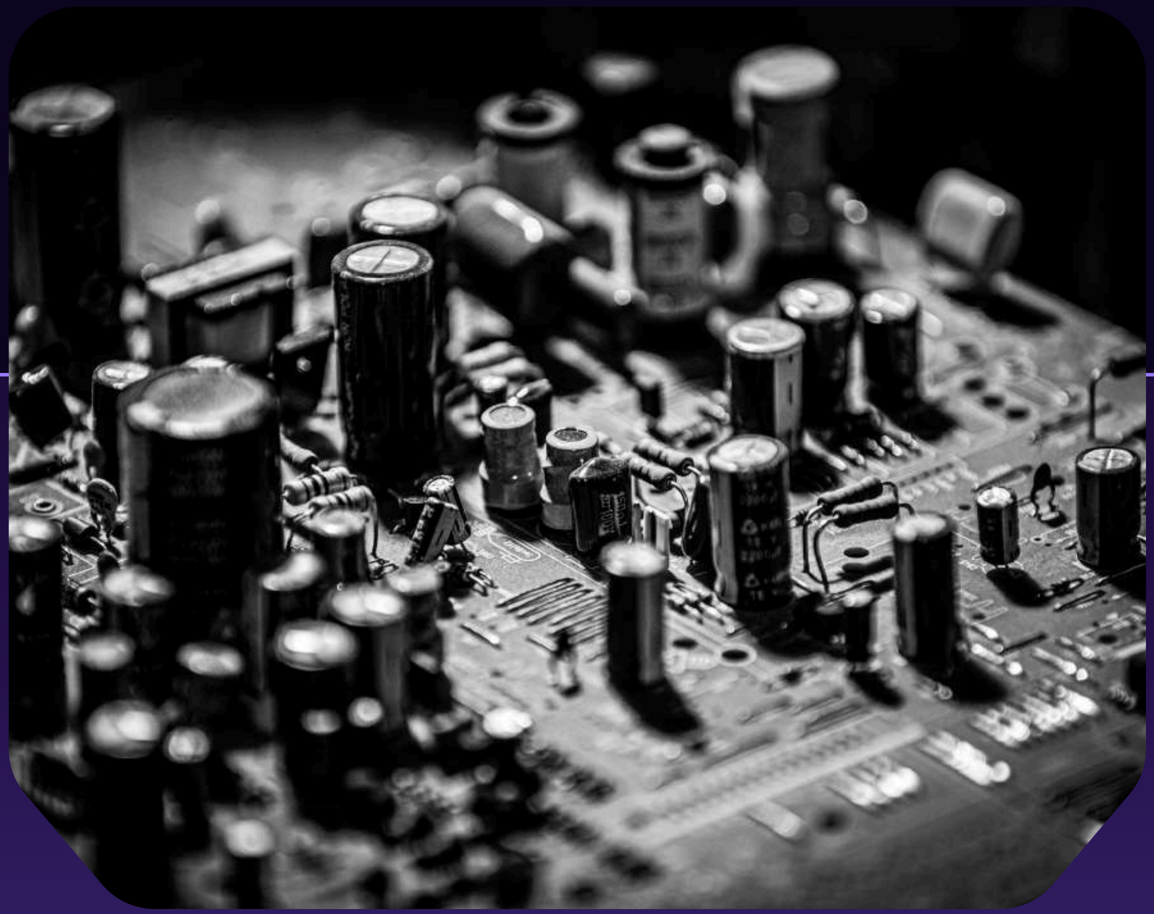
BOTTOM UP APPRACH

Sol-Gel Method

Pros: Exceptional purity and chemical homogeneity; cost-effective for fabricating complex nanostructures.

Cons: A time-consuming (tedious) process; potential health hazards due to organic chemical use; mandatory post-treatment purification.

Giggling Platypus Co.



MAIN FEATURES

TYPES OF NANO MATERIALS

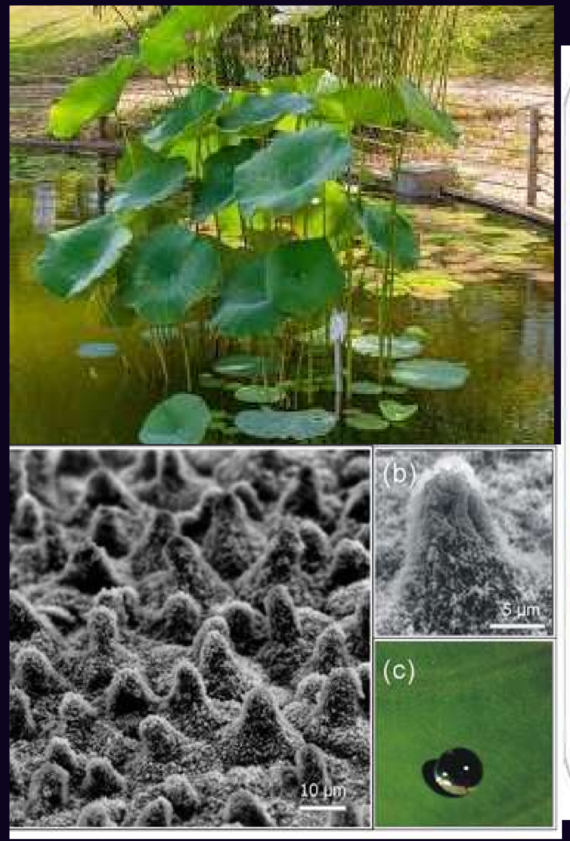
TYPES OF NANO MATERIALS

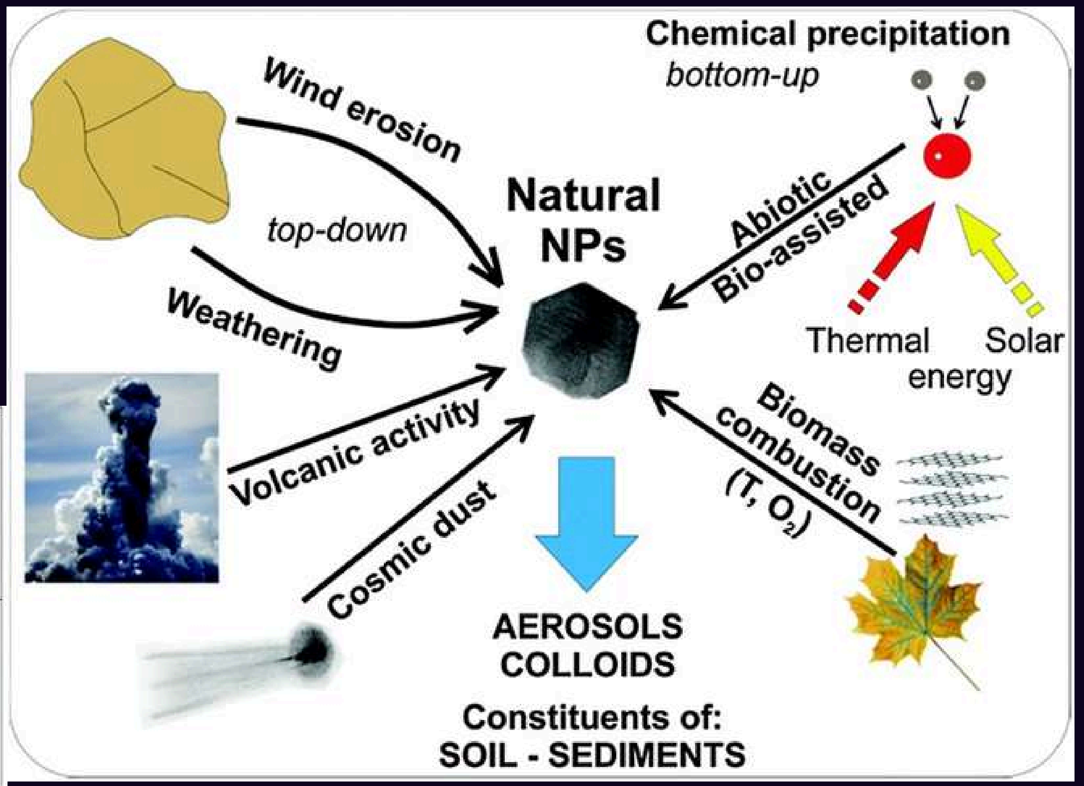
BASED ON ORIGIN

natural/artificial

Natural Nanomaterial

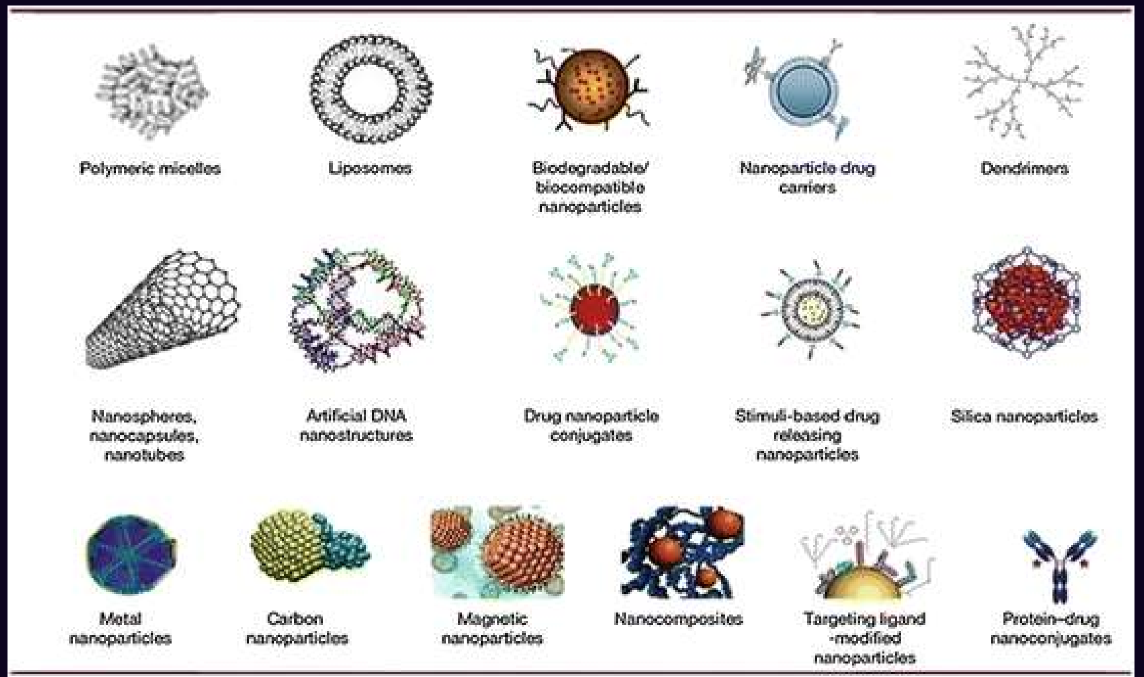
Found in nature in various forms , such as viruses, protein molecules (e.g., ferritin), minerals like clay, and biological structures like lotus leaves or volcanic ash (geological process)





Artificial(Engineered) Nanomaterial

Nanomaterials that are synthesized using precise mechanical and manufacturing procedures.
Designed at the atomic or molecular level to achieve specific physical and chemical properties.

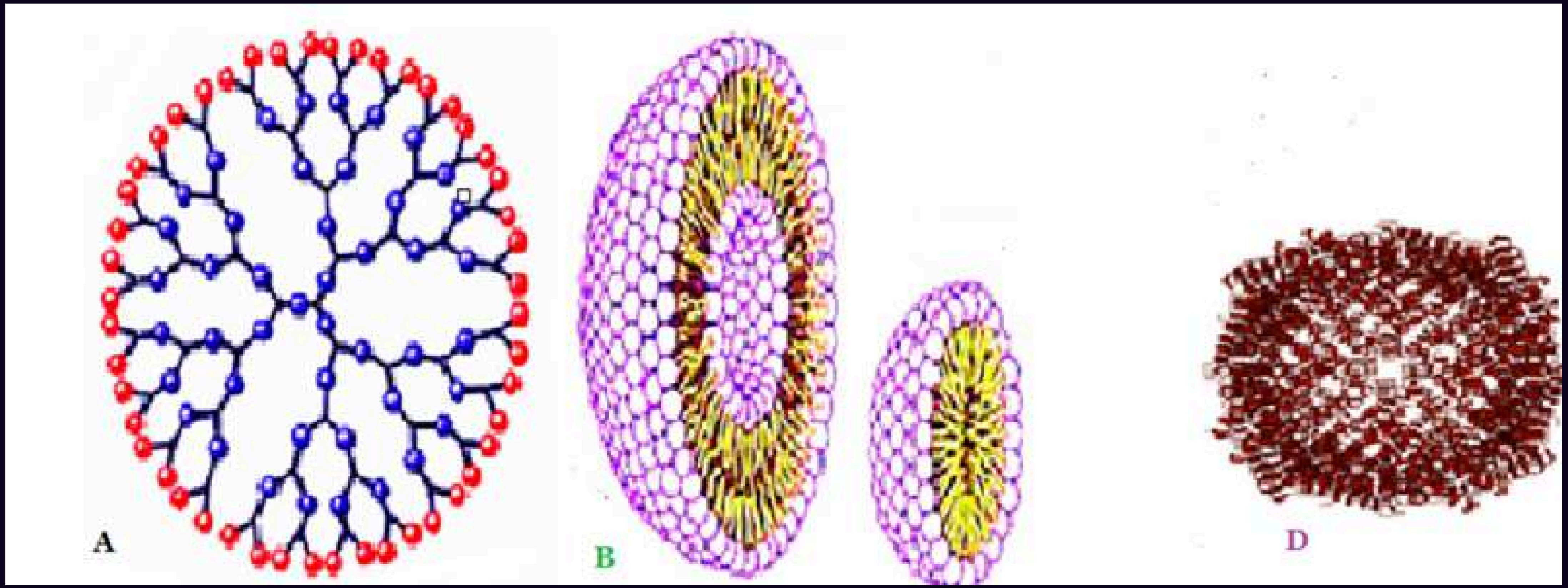


TYPES OF NANO MATERIALS

BASED ON STRUCTURAL CONFIGURATION/ COMPOSITION

orgnic/inorgnic/crbon bsed/composite

ORGANIC NANOMATERIALS



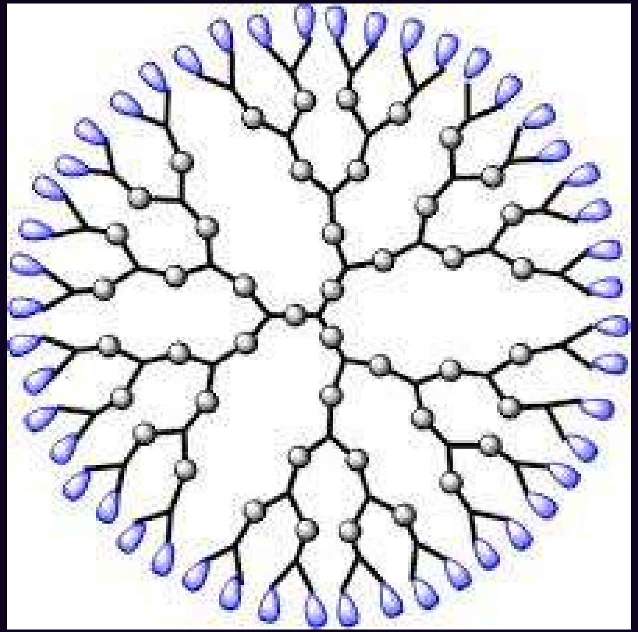
-Primarily made of organic molecules (lipids, polymers, or biopolymers).
-Non-toxic, biodegradable, and *biocompatible (frequently used in the process of transporting target medications to their intended locations)*
-Capable of being broken down naturally (Eco-friendly).
Highly sensitive to thermal (heat) and photo (light) stimuli—ideal for "triggered" release.

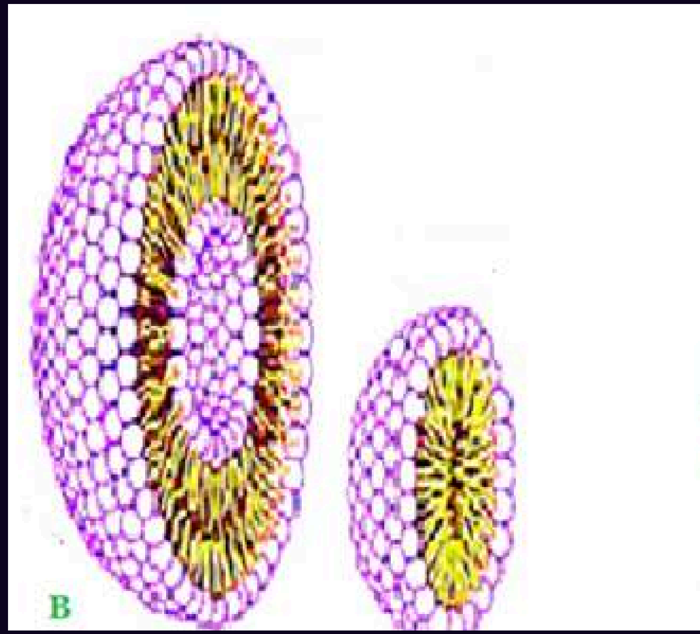
TYPES OF NANO MATERIALS

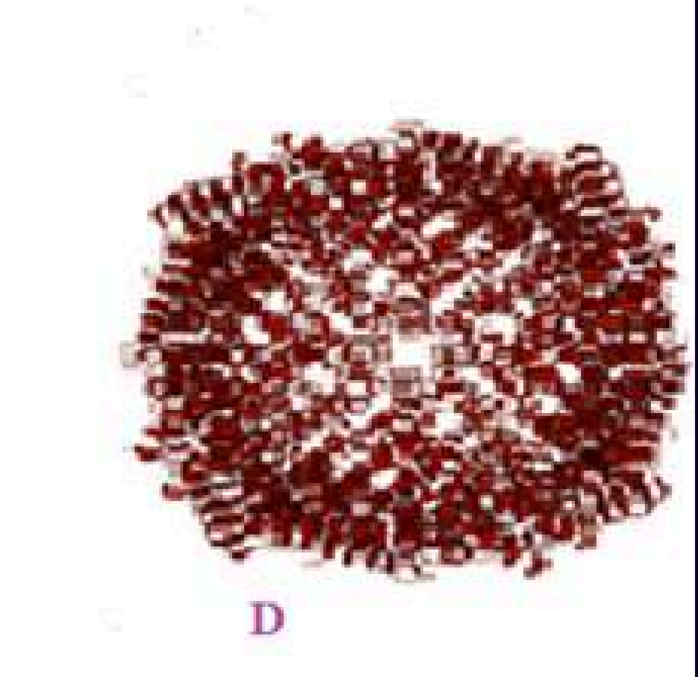
BASED ON STRUCTURAL CONFIGURATION/ COMPOSITION

organic/inorganic/carbon based/composite

ORGANIC NANOMATERIALS







A. Dendrimers- highly branched, ***tree-like synthetic polymers*** with a well-defined, three-dimensional structure.

B. Liposomes-**spherical** vesicles composed of one or more phospholipid bilayers, similar to cell membranes.

C. Micelles -self-assembled spherical structures formed by **amphiphilic molecules** in aqueous solutions.

D. Ferritin- natural protein nanocage that stores and releases iron in biological systems.

TYPES OF NANO MATERIALS

BASED ON STRUCTURAL CONFIGURATION/ COMPOSITION organic/inorganic/carbon based/composite

INORGANIC NANOMATERIALS

-nanoparticles that lack car bon atoms

Metal Based Nano particles

Nanoparticles derived from metals such as:

- Aluminum (Al), Cadmium (Cd), Cobalt (Co), Copper (Cu)
- Gold (Au), Iron (Fe), Lead (Pb), Silver (Ag)-, Zinc (Zn)

Synthesized through:

- Destructive processes
- Constructive processes

Properties:

- High surface-to-volume ratio
- Strong quantum effects
- Excellent UV-Visible sensitivity
- Enhanced electrical, thermal, catalytic, and antibacterial properties

Applications:

- Electronics, medicine, catalysis, sensors, and antibacterial materials

TYPES OF NANO MATERIALS

BASED ON STRUCTURAL CONFIGURATION/ COMPOSITION organic/inorganic/carbon based/composite

INORGANIC NANOMATERIALS

-nanoparticles that lack car bon atoms

Metal Oxide Nanoparticles

Composed of Positive metal ions and negative oxygen ions

Examples:
- Silicon dioxide (SiO_2), Titanium dioxide (TiO_2)
- Zinc oxide (ZnO), Aluminum oxide (Al_2O_3)

Properties:
- More stable than pure metal nanoparticles
- Distinct electrical, optical, and catalytic properties
- Useful in environmental and industrial applications

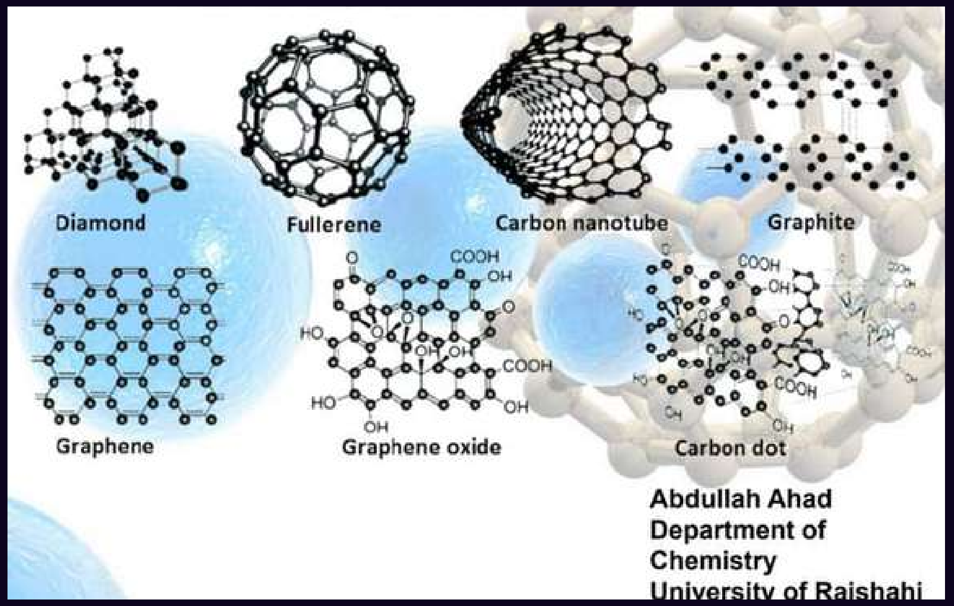
TYPES OF NANO MATERIALS

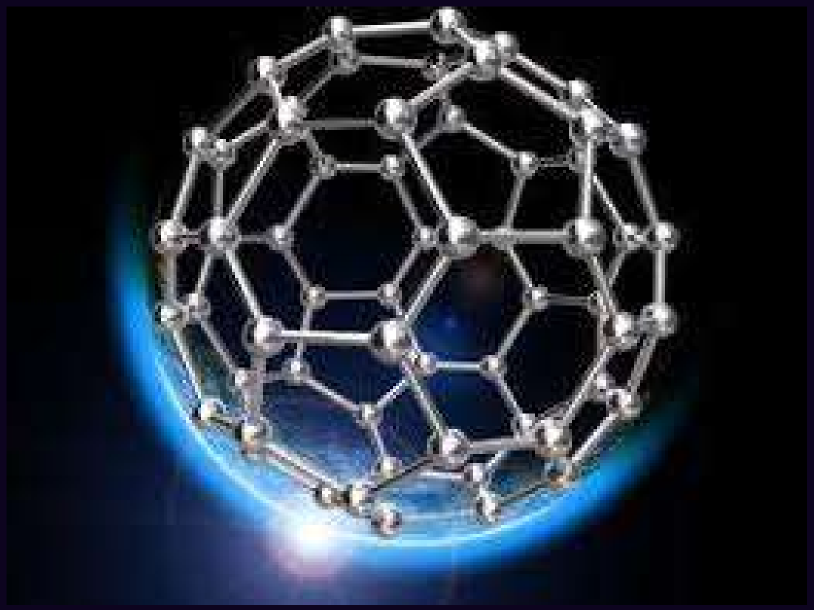
BASED ON STRUCTURAL CONFIGURATION/ COMPOSITION

orgunic/inorgunic/curbon bused/composite

CARBON BASED NANOMATERIALS

Formed entirely of carbon atoms arranged in various allotropes- different structural forms of the same element in the same physical state (solid, liquid, or gas), possessing distinct physical and chemical properties.
categorized primarily by their geometric configuration and structural lattices. They are recognized for being stronger than steel and possessing unique thermal properties: (anisotropic--conductive along the length, non-conductive across).





Classification of CNM's

1. Fullerenes (Buckyballs)

- Structure: Spherical or ellipsoidal configurations of carbon atoms.
- Dimensions: Single-layer: Up to 8.2 nm in diameter.
  - Multi-layered: 4 to 36 nm in diameter.
- Composition: Composed of 28 to 1500 carbon atoms arranged in a hollow cage.

TYPES OF NANO MATERIALS

BASED ON STRUCTURAL CONFIGURATION/ COMPOSITION organic/inorganic/carbon based/composite

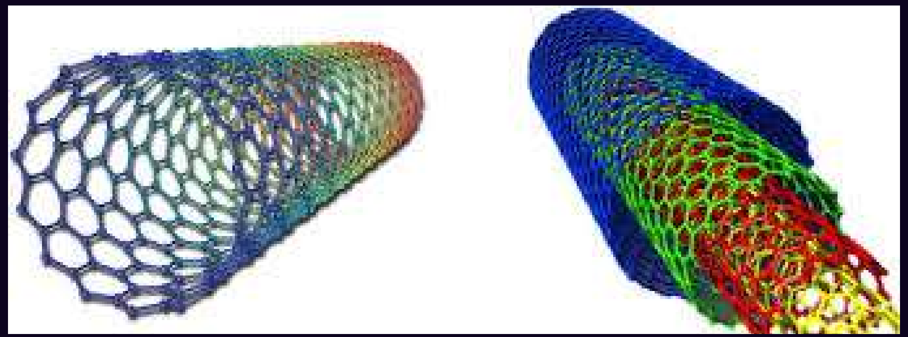
Classification of CBNs

2.Graphene

Structure: A two-dimensional (2D) planar sheet of carbon atoms arranged in a hexagonal honeycomb lattice.

Dimensions: The thickness of a single sheet is approximately 1 nm.

Significance: It serves as the fundamental building block for other carbon allotropes (e.g., it can be wrapped into nanotubes or stacked into graphite).



3. Carbon Nanotubes (CNTs)

- Structure: Hollow cylinders formed from rolled graphene sheets.

Types:

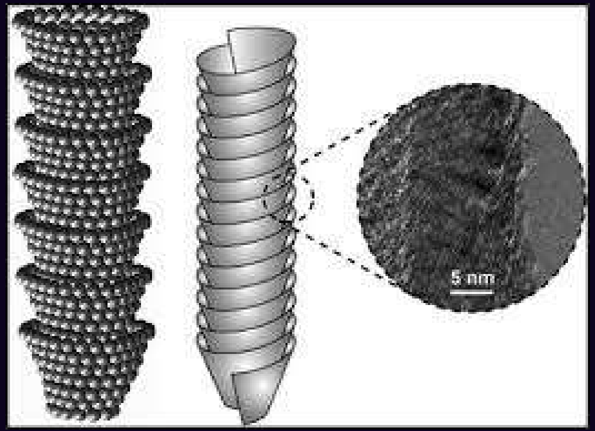
- Single-Walled (SWCNTs): Diameters as low as 0.7 nm.
- Multi-Walled (MWCNTs): Diameters up to 100 nm.
- Scale: Lengths vary significantly from a few micrometers to several millimeters.

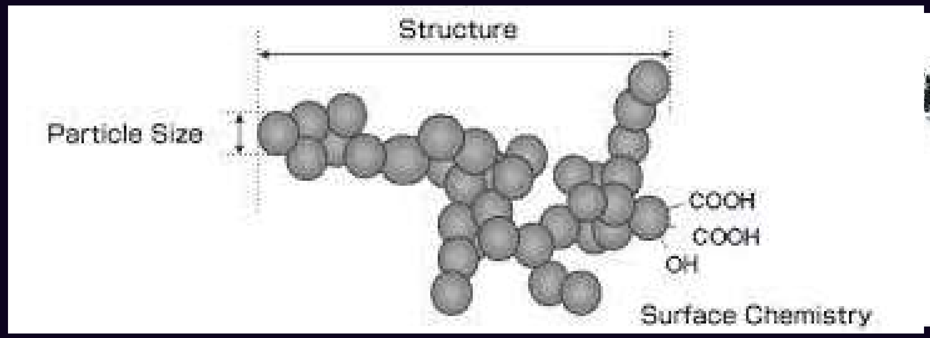
TYPES OF NANO MATERIALS

BASED ON STRUCTURAL CONFIGURATION/ COMPOSITION

organic/inorganic/carbon based/composite

Classification of CBNs





4. Carbon Nanofibers (CNF)

- Structure: Produced from graphene nano-fossils; these are cylindrical nanostructures with graphene layers arranged in specific stacking patterns (cones, cups, or plates).
-
- Properties: Widely used for structural reinforcement due to high aspect ratios and mechanical strength.

5. Carbon Black

- Structure: An amorphous, generally spherical material.
- Dimensions: Diameters range from 20 to 70 nm.
- Use Case: Often used as a pigment and reinforcing phase in automobile tires and plastic products.
-

TYPES OF NANO MATERIALS

BASED ON STRUCTURAL CONFIGURATION/ COMPOSITION organic/inorganic/carbon based/composite

COMPOSITE NANOMATERIALS

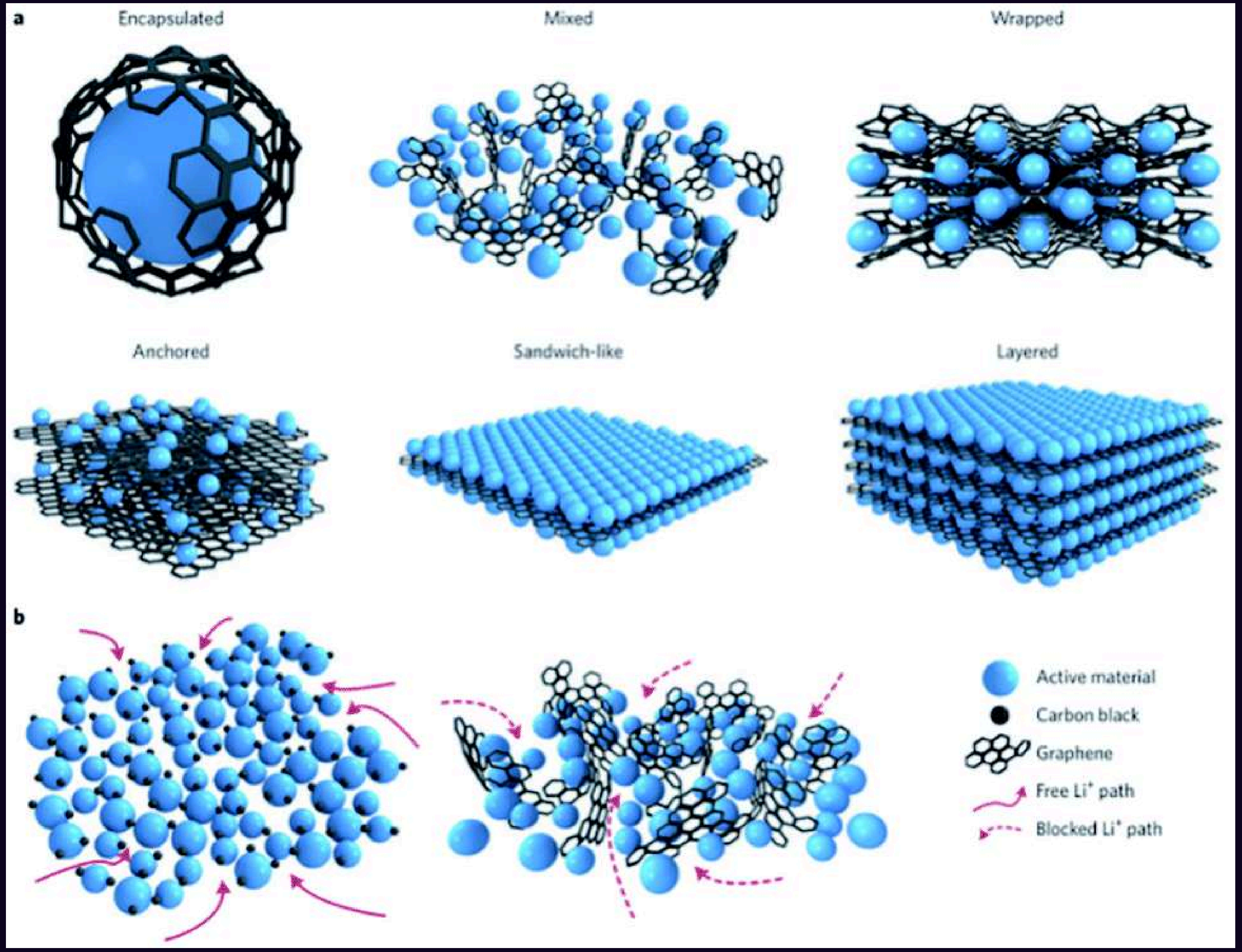
complex materials made by combining nanoparticles with other nanoparticles or bulk-scale materials to improve mechanical or thermal properties.

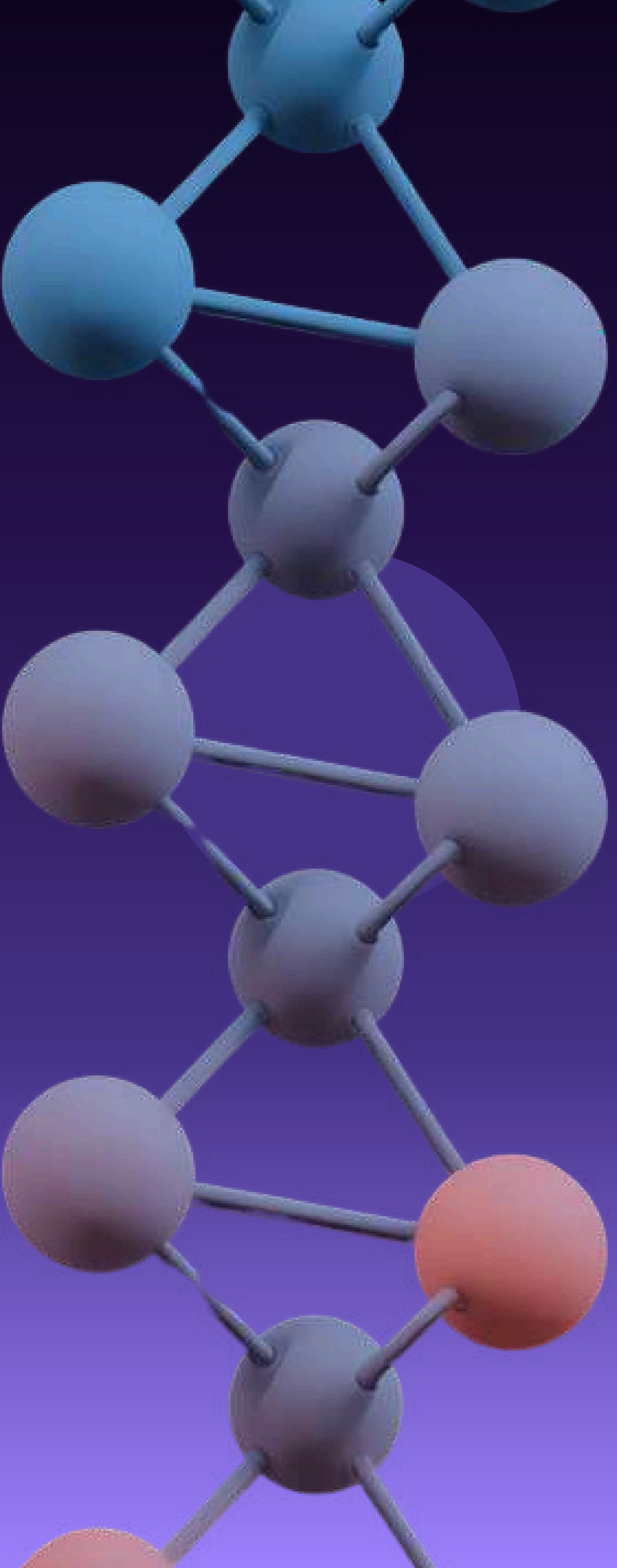
Nanoparticle + Nanoparticle

- Silver and Iron Oxide
- hybrid with dual properties (like antimicrobial and magnetic).

-Enhanced Properties

- -Mechanical Strength
- Thermal Stability
- Flame Retardancy





EXAMPLES AND APPLICATIONS ACROSS FIELDS

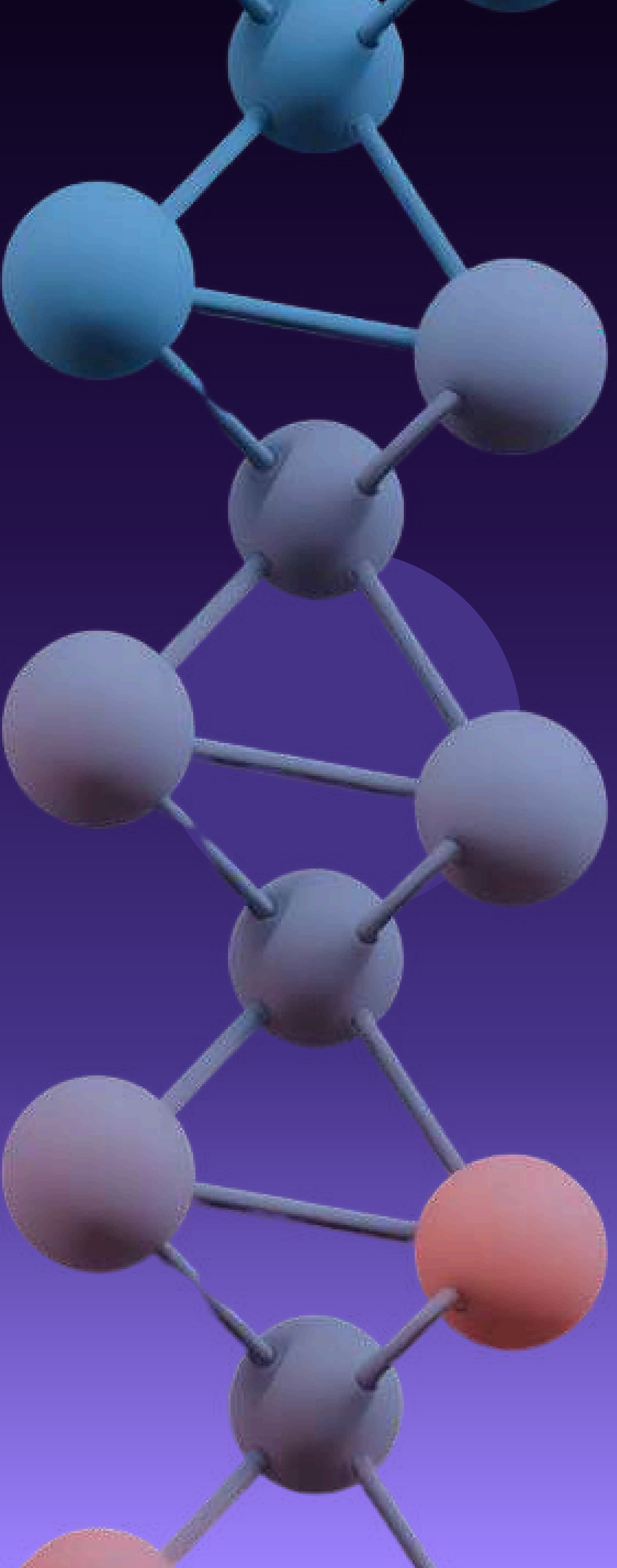
KEY EXAMPLES OF NANOMATERIALS

Carbon Nanotubes: Cylindrical carbon structures with exceptional mechanical strength and electrical conductivity.

Quantum Dots: Semiconductor nanoparticles with size-tunable optical properties for displays and imaging.

Nanoclusters: Small aggregates of atoms exhibiting distinct catalytic behavior and unique reactivity patterns.

Fullerenes: Spherical carbon molecules such as C60, offering applications in drug delivery and materials science.



CARBON NANOTUBES

STRUCTURE OF NANOTUBES

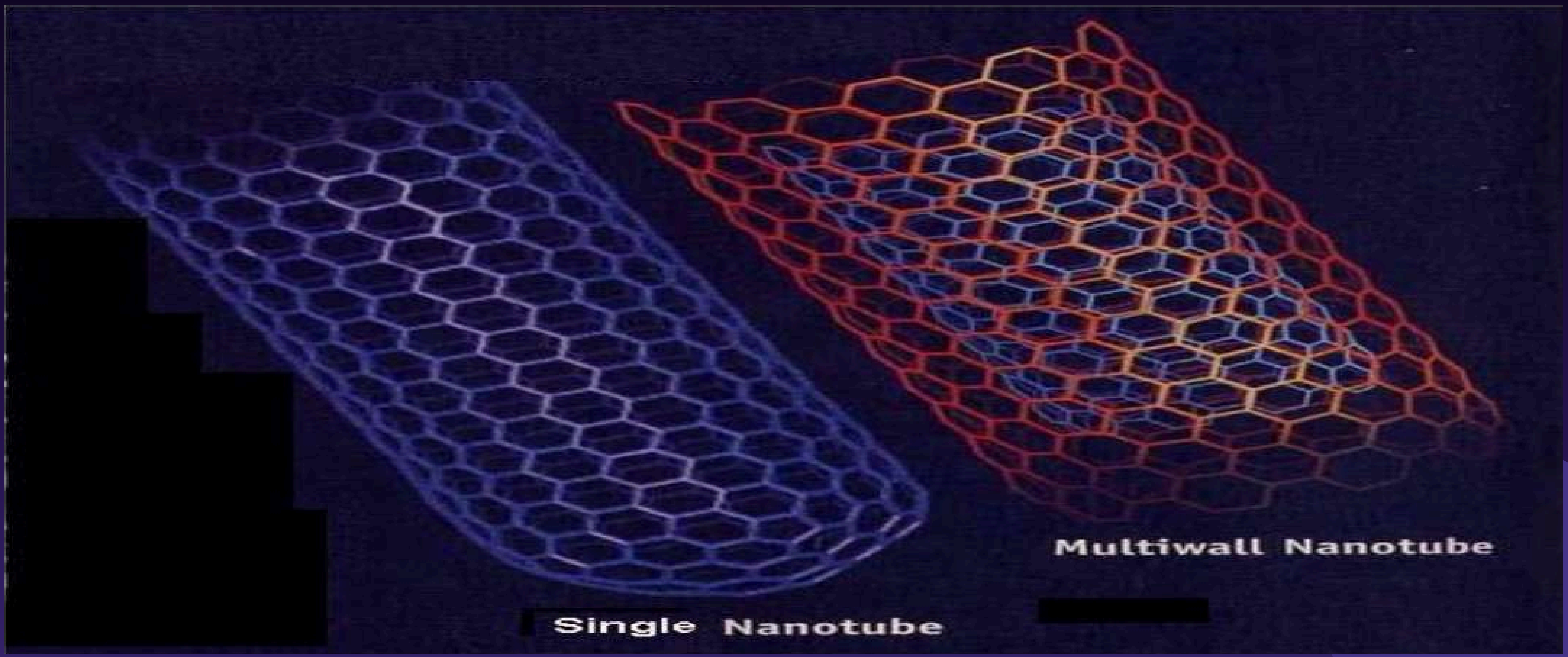
A carbon nanotube is a tube shaped material, made of carbon and has diameter on the nanometer scale. Two types of carbon nanotubes are observed Single walled nanotubes (SWNT) and Multiwalled nanotubes (MWNT).

SINGLE WALLED NANOTUBES (SWNT):

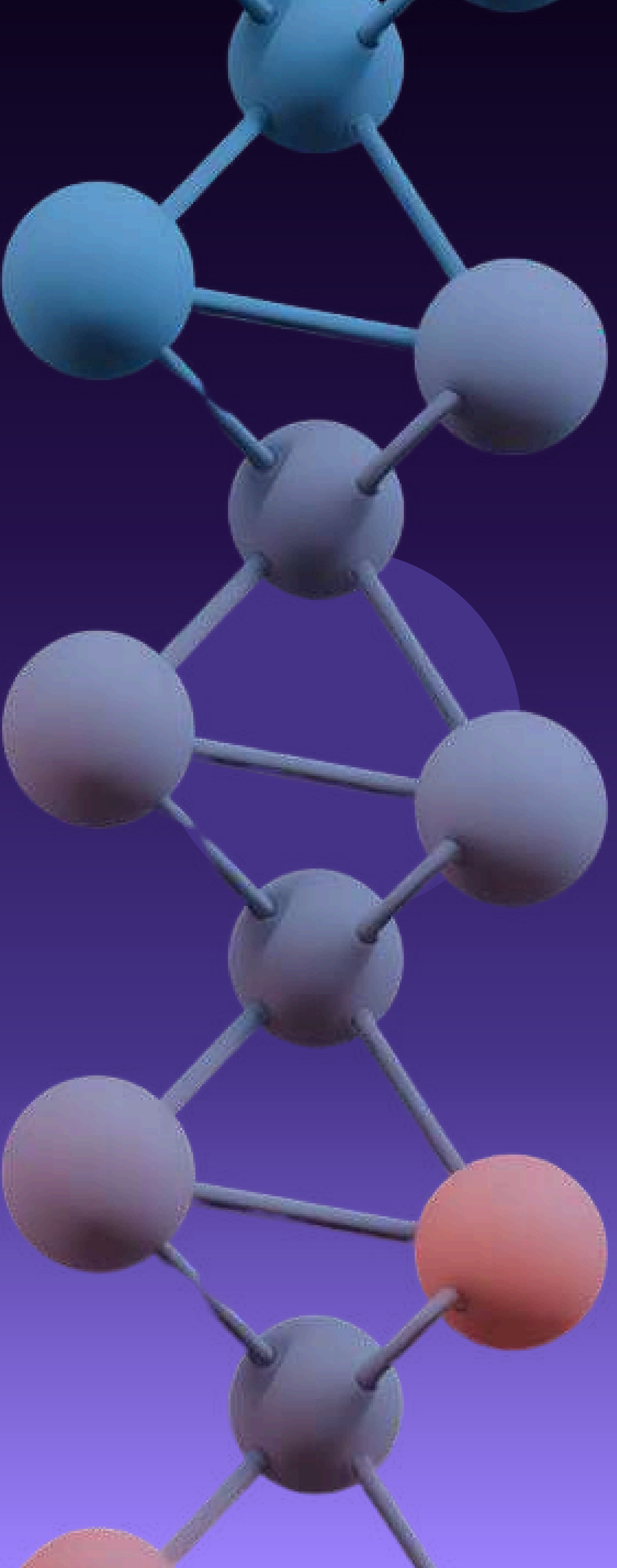
They have diameter close to 1 nanometer and tube length may vary millions of times longer. They can be formed in three different forms – archair, zigzag and chiral forms on basis of rolling of grapheme sheet in to a seamless cylinder. Each form has a special effect on electrical property of nanotube.

MULTIWALLED NANOTUBES (MWNT):

The discovery of C60 motivated the researchers to search for other carbon compounds containing curved graphenes. This led to the discovery of multiwalled nanotubes were made of concentric cylinders of rolled-up graphene sheets, capped with semi-fullerenes. The length of a tube was in the range of a few μm, the diameter was 10-20 nm. As the size increases, these structures exhibit properties between fullerenes and graphite.



Two models can be used to describe the structure of multiwalled nanotubes. In Russian doll model, sheets of graphite are arranged in concentric cylinders within a larger single walled nanotube. In the parchment model, single sheet of graphite is rolled in around itself, resembling a scroll of parchment or rolled newspaper. The interlayer distance in multiwalled nanotubes is close to distance between graphene layers approximately 3.3Å. They have greater tensile strength than single walled nanotubes.



QUANTUM DOTS

Quantum dots are small regions defined in the semiconductor materials with the same size of the distance in an electron-hole pair. The physics of quantum dots has been a very active and fruitful research topic. Their unique optical, photochemical, semiconductor, and catalytic properties are due to the quantum confinement.

To date, chemistry, physics, and materials science have provided methods for the production of quantum dots and allow tighter control of factors affecting, for example, particle growth and size, solubility and emission properties. This book deals with the electronic and optical properties of quantum dots as an artificially fabricated device.

June 2020
DOI: 10.5772/intechopen.92151
In book: Quantum Dots - Fundamental and Applications

QUANTUM DOTS

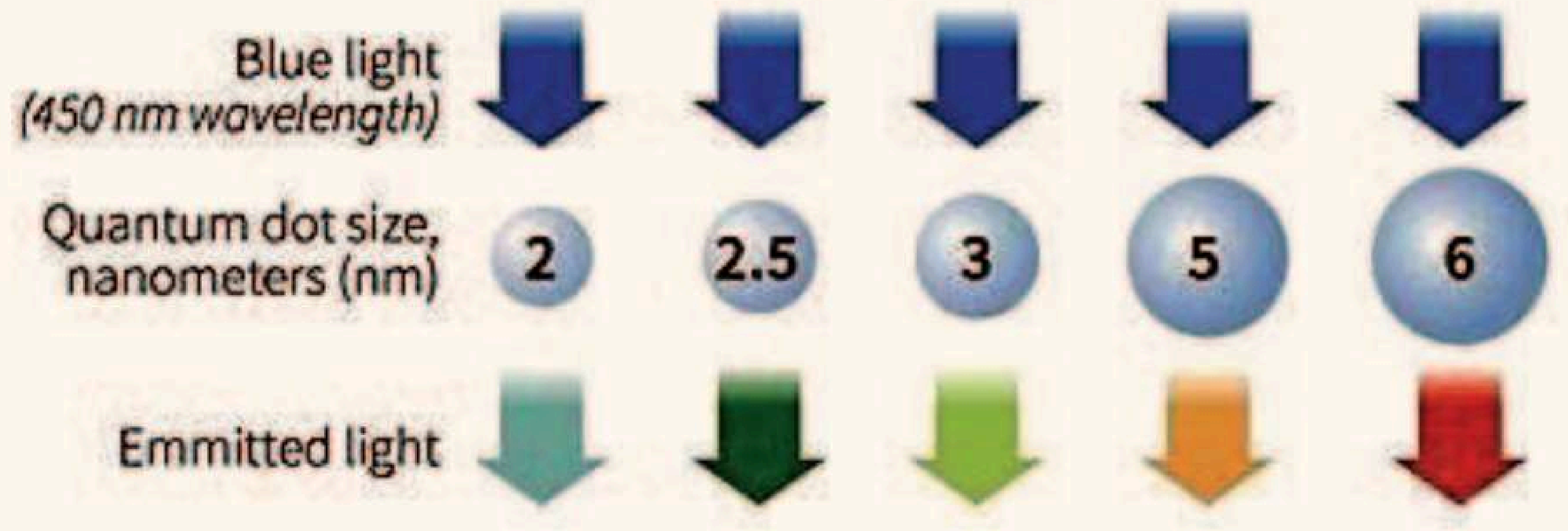
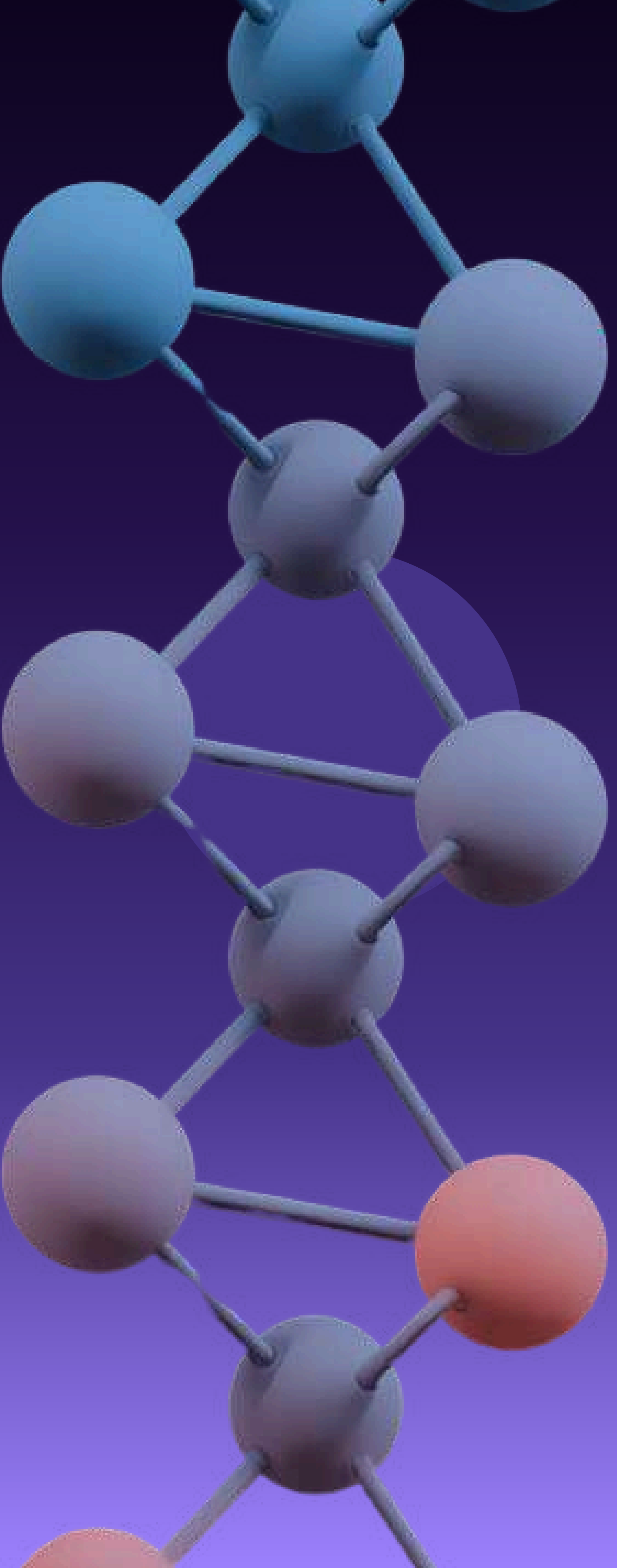


Figure 1.
Conversion of the light spectrum into different colors depends on the quantum dot size (image: RNGS Reuters/ Nanosys).

As shown in the Figure, different sized quantum dots emit different color light due to quantum confinement. The physics of quantum dots counterpart with a lot of the behaviors that naturally occurring in atomic and nuclear physics of the quantum system.

June 2020
DOI: 10.5772/intechopen.92151
In book: Quantum Dots - Fundamental and Applications

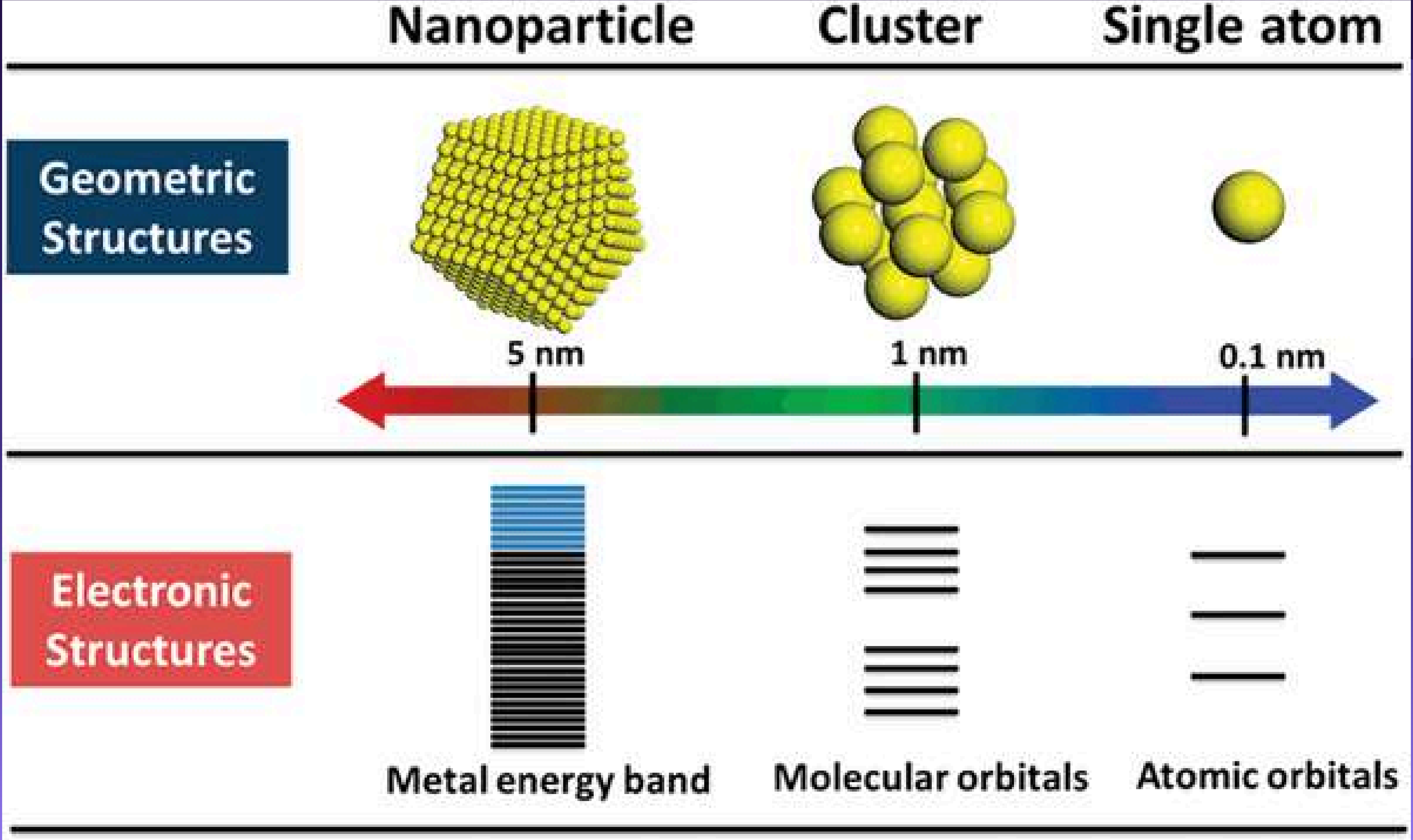


NANOCLUSTERS

Clusters are aggregates of atoms or molecules, and they can be formed at any size and from any substance. There are two main purposes for studying clusters; (a) for understanding the evolution of materials properties from the atomic/molecular/microscopic world to that of the bulk/condensed matter/macroscopic world and (b) for exploring the novel properties exhibited by some cluster sizes and thus developing the possibility of applications. The term, nanoclusters, simply refers to large clusters whose dimensions are most conveniently measured in nanometers. An outline of the contents of our presentation is provided by its title. There is a molecular beam part and a bulk materials pan. with each of these further divided into sections on generation and charaterization.

NANOCLUSTERS

STRUCTURE



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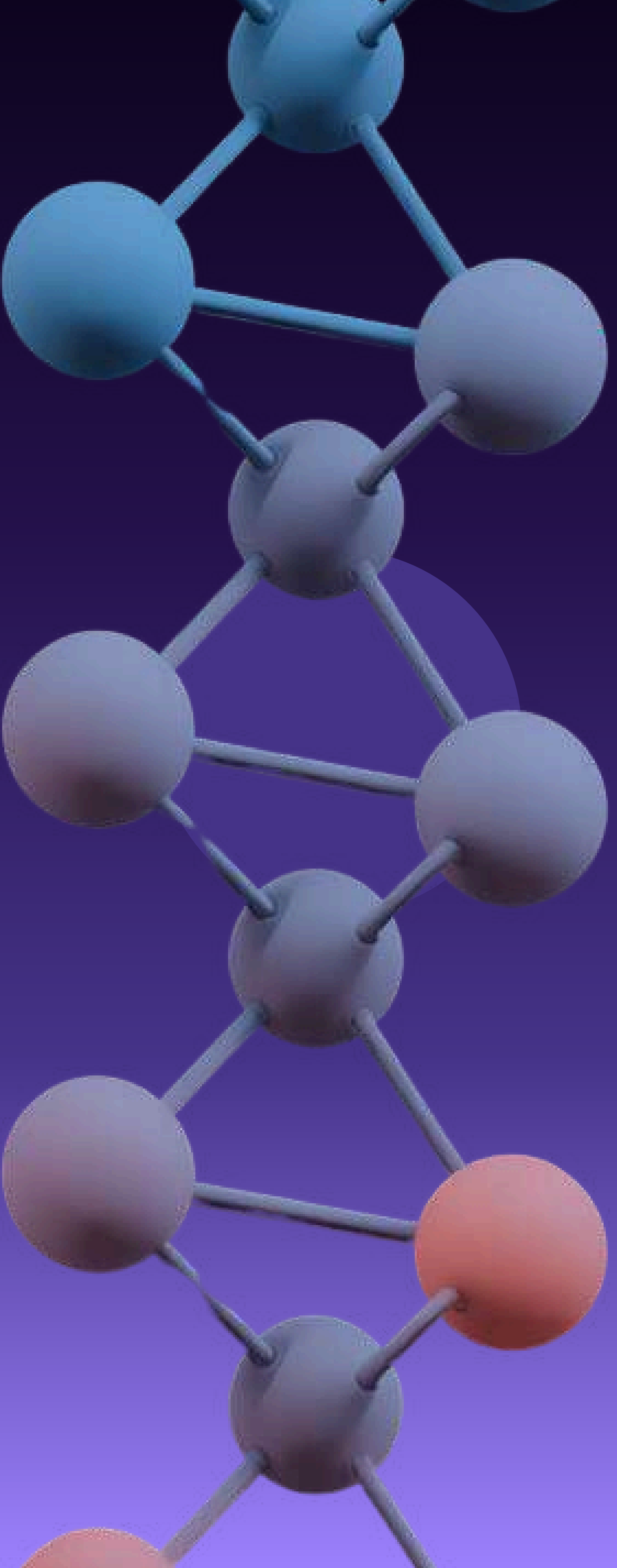
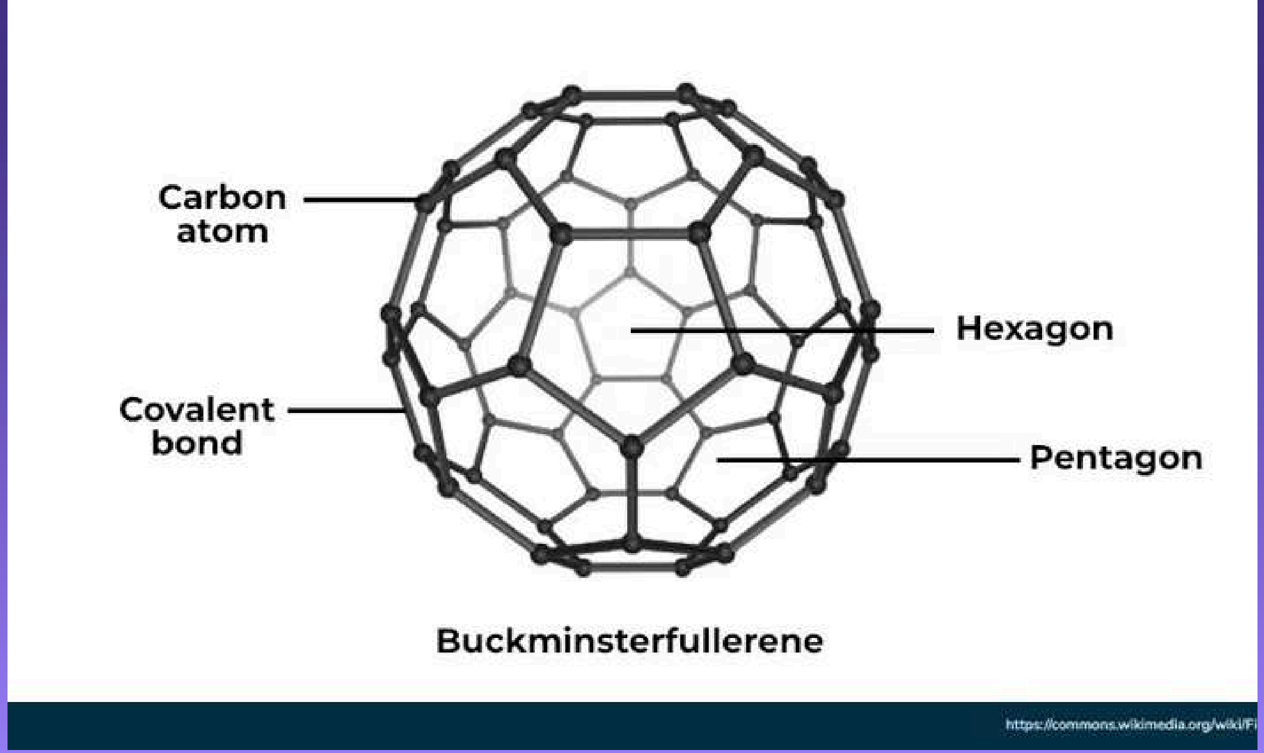


FULLERENES

A fullerene is a carbon cage structure having fused ring system which consists of pentagons and hexagons. The first proposal of buckyball was given by Eiji Osawa, Japan. He recognized that corannulene, a cyclopentane ring fused with 5 benzene rings was a part of football framework and hypothesized that entire structure could exist. In 1985, the group of scientists Richard Smalley, Robert Curl, James Heath, Sean O'Brien, and Harold Kroto synthesized the first fullerene molecule, buckminsterfullerene (C60) at Rice University. These scientists named newly discovered molecule in honor of architect R. Buckminster Fuller who created geodesic dome with same shape.



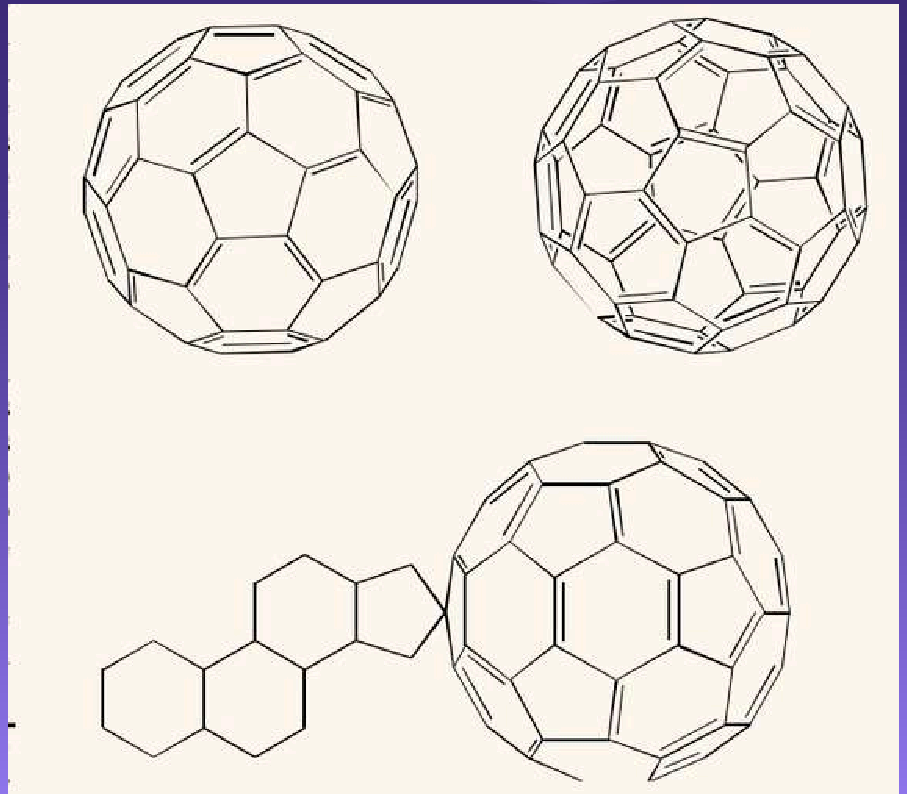
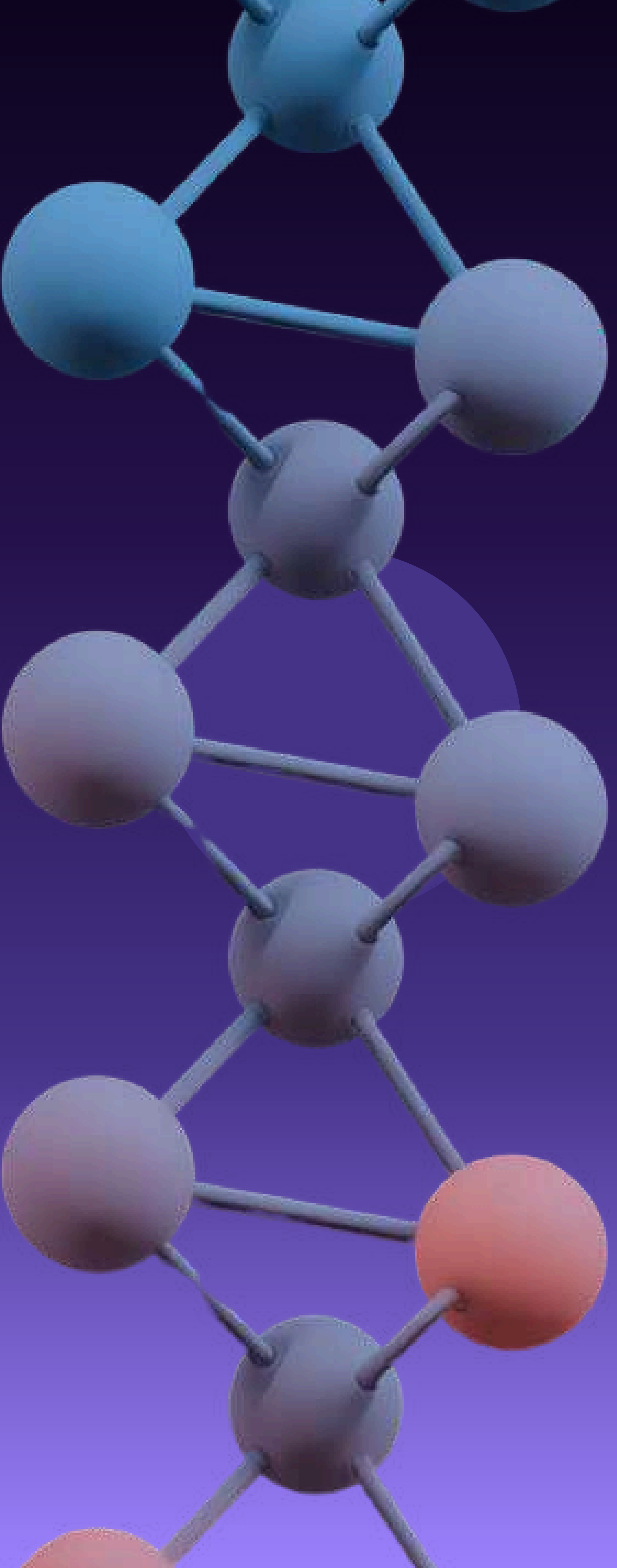
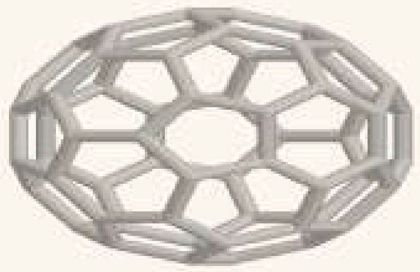
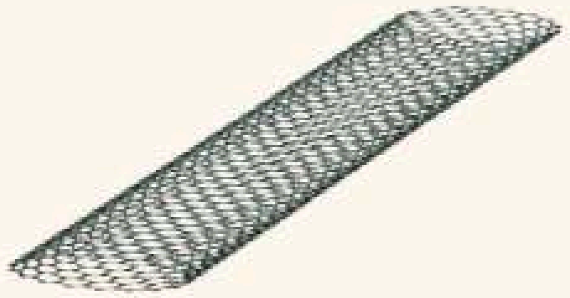
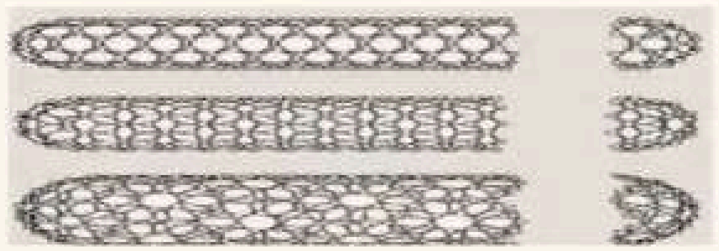


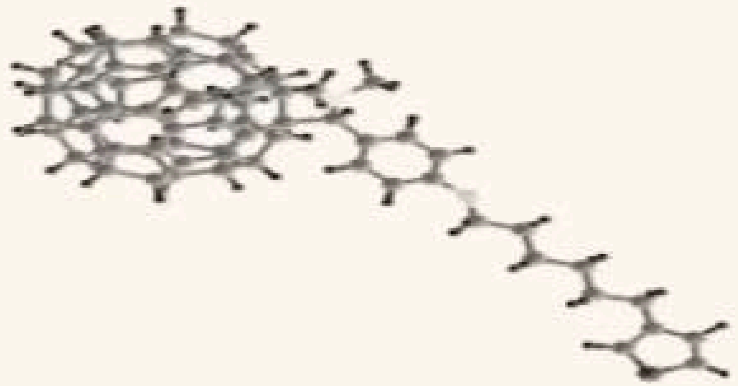
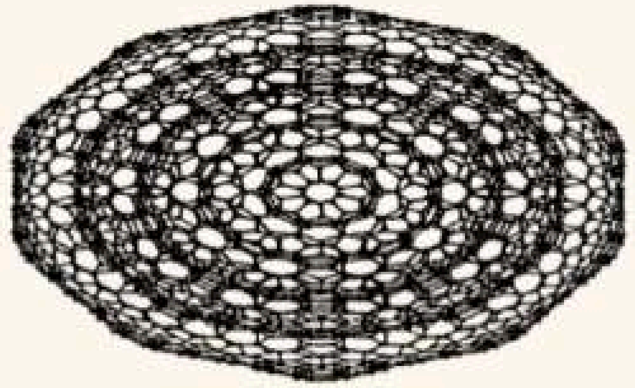
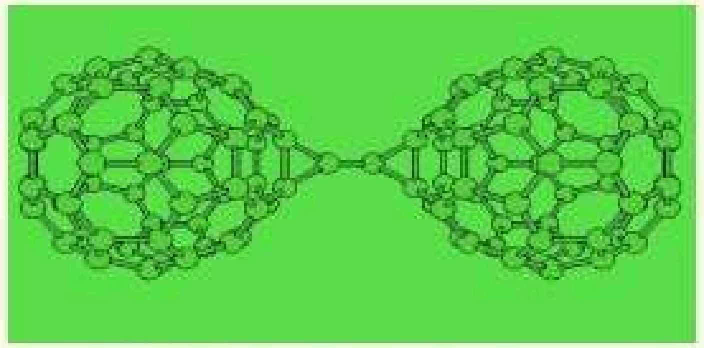
Fig. (1). Different fullerene derivatives.



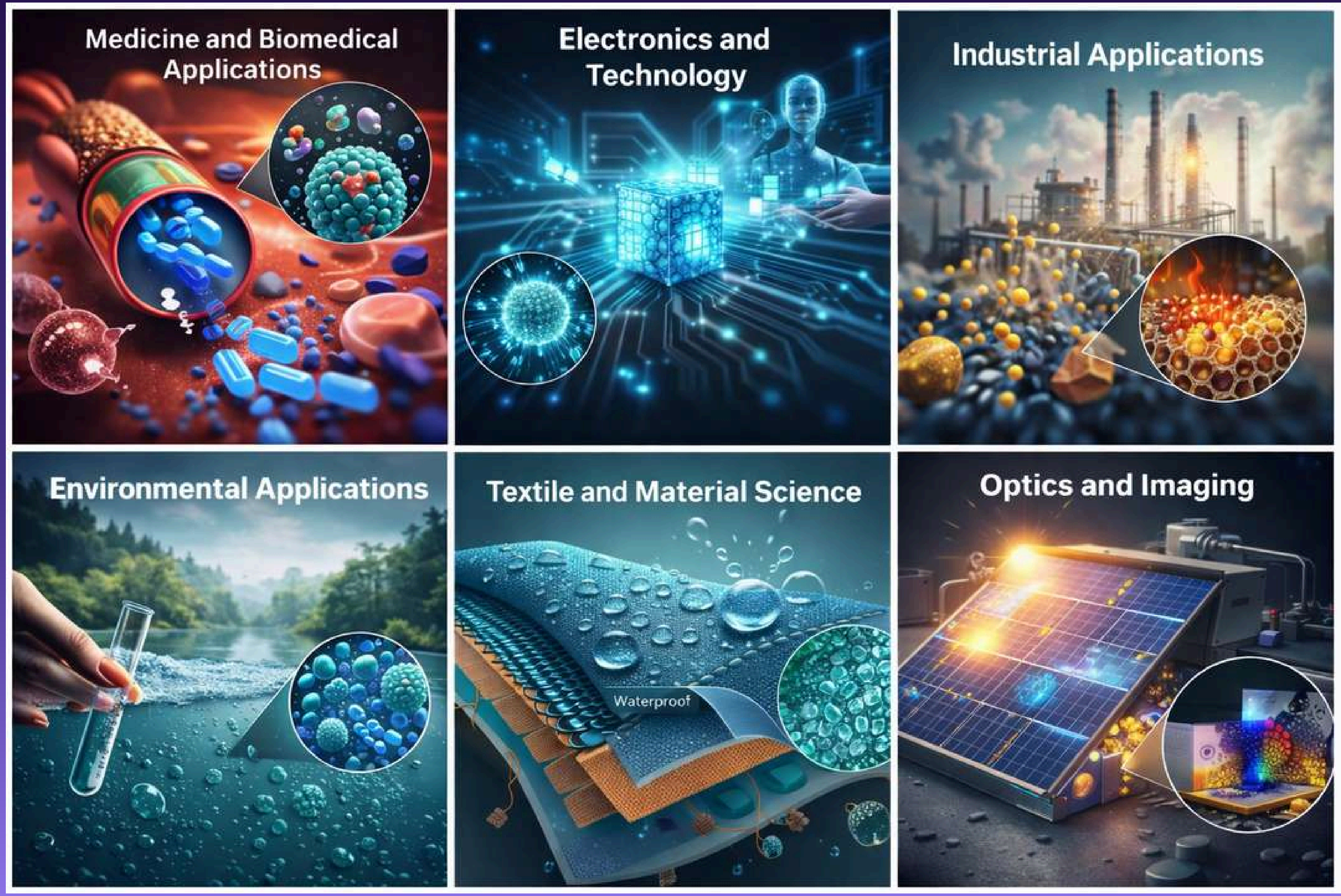
2. TYPES OF FULLERENES

Table 1. Types of fullerenes.

Sr. no.	Types	Description	Representation
1.	Buckyball clusters [4] (Fig. **2**)	Smallest number is C_{20} and most common is C_{60}. Because sphere of buckyball is hollow, other atoms can trapped within it.	 **Fig. (2).** Buckyball clusters.
2.	Nanotubes [5] (Fig. **3**)	Hollow tubes of very small dimensions with single or multiple walls, used in electronics industry.	 **Fig. (3).** Nanotubes.
3.	Mega tubes [6] (Fig. **4**)	Varying in dimensions than nanotubes as larger in diameter with walls of different thickness; application in transport of a variety of molecules of different sizes.	 **Fig. (4).** Mega tubes.

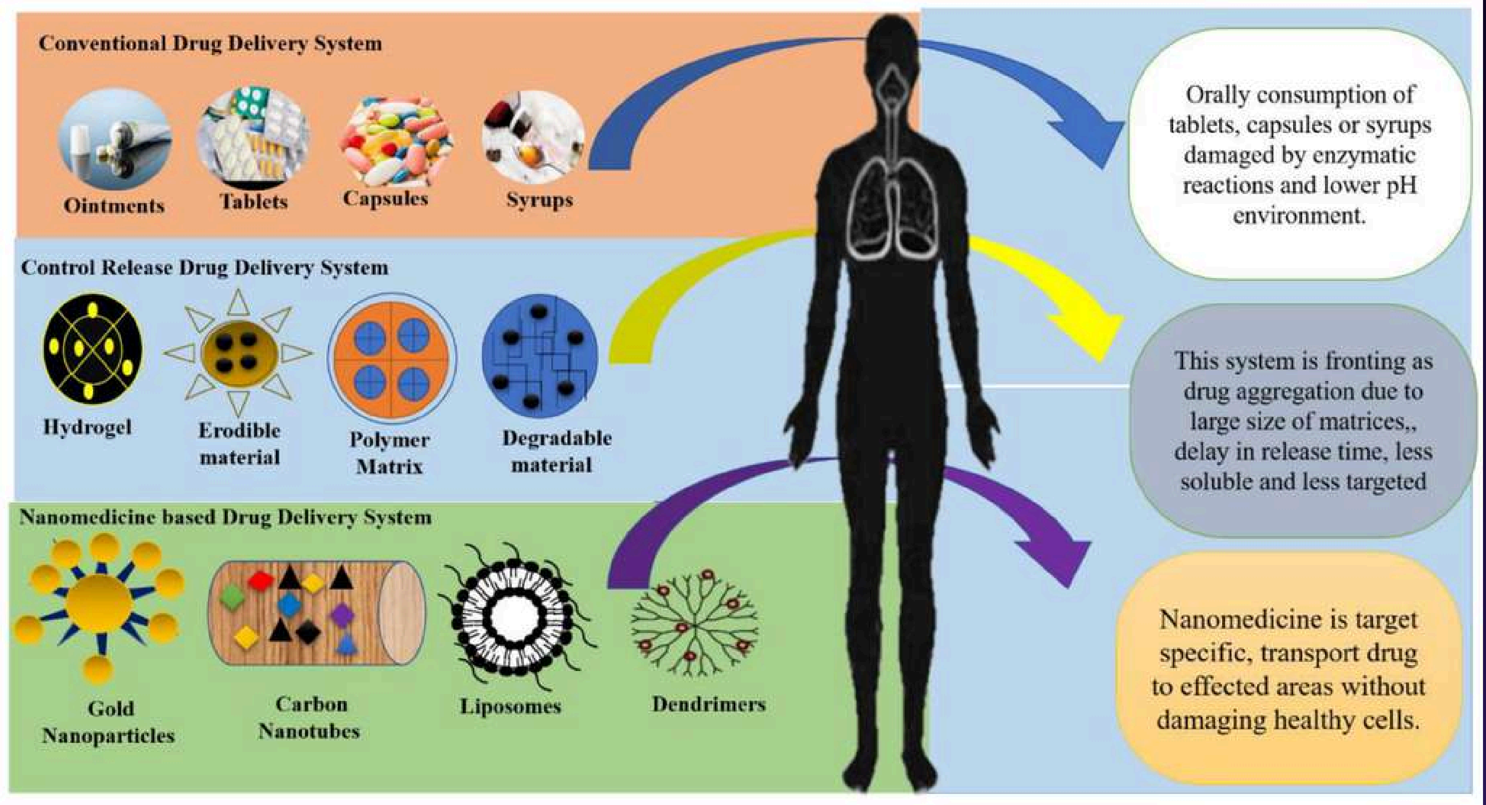
4.	Polymers [7] (Fig. **5**)	Chain, two or three dimensional, formed under high pressure and temperature conditions.	 **Fig. (5).** Polymers.
5.	Nano"onions" [8] (Fig. **6**)	Spherical particles having multiple layers which surrounds buckyball core. Typical diameter is 3-5 nm; proposed for lubricants.	 **Fig. (6).** Nano"onions".
6.	Linked "ball-and-chain" dimers [9] (Fig. **7**)	Two buckyballs linked by a carbon chains.	 **Fig. (7).** Linked ball and chain dimers.

APPLICATIONS ACROSS FIELDS



APPLICATIONS OF NANOCHEMISTRY IN DIFFERENT FIELDS

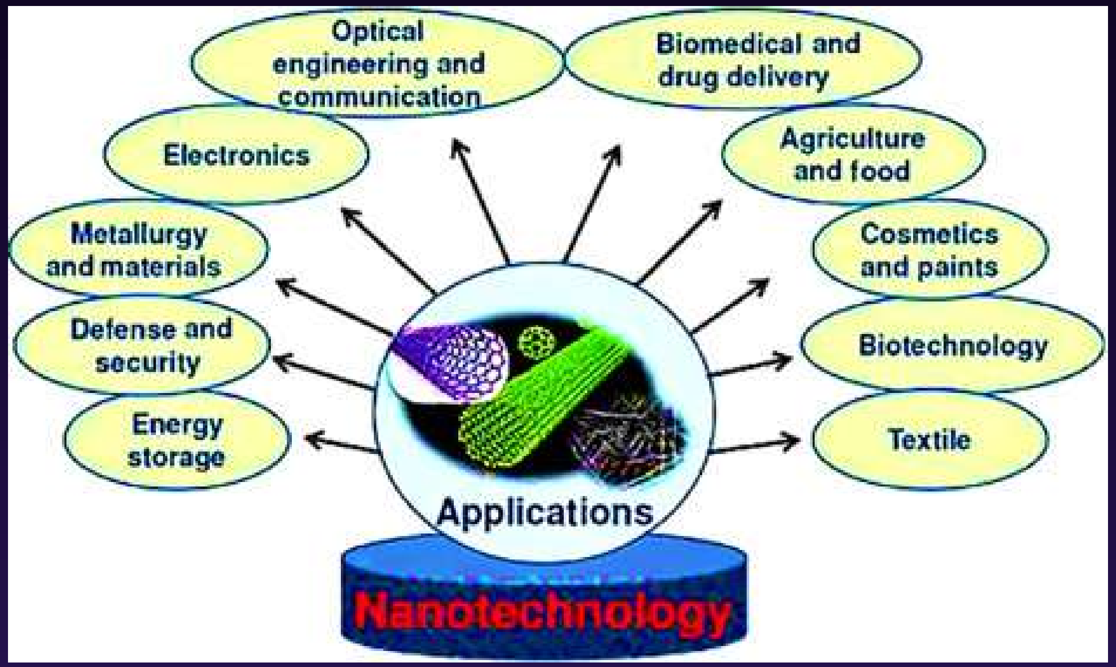
Nanochemistry plays a vital role in modern science and technology due to the unique properties of materials at the nanoscale. These materials exhibit enhanced chemical, physical, and biological properties, making them useful across multiple fields.



MEDICINE AND BIOMEDICAL APPLICATIONS

- Drug Delivery Systems
  - Nanoparticles are used as carriers to deliver drugs directly to target cells, improving effectiveness and reducing side effects.
- Cancer Treatment
  - Nanocarriers can transport anticancer drugs specifically to tumor cells, improving treatment efficiency.
- Medical Imaging
  - Nanoparticles are used in imaging techniques such as MRI for better diagnosis.
- Gene Therapy
  - Nanotechnology enables the delivery of DNA into cells for treating genetic disorders and diseases.
- Antibacterial and Infection Treatment
  - Nanoparticles enhance antibiotic effectiveness and help treat intracellular infections.

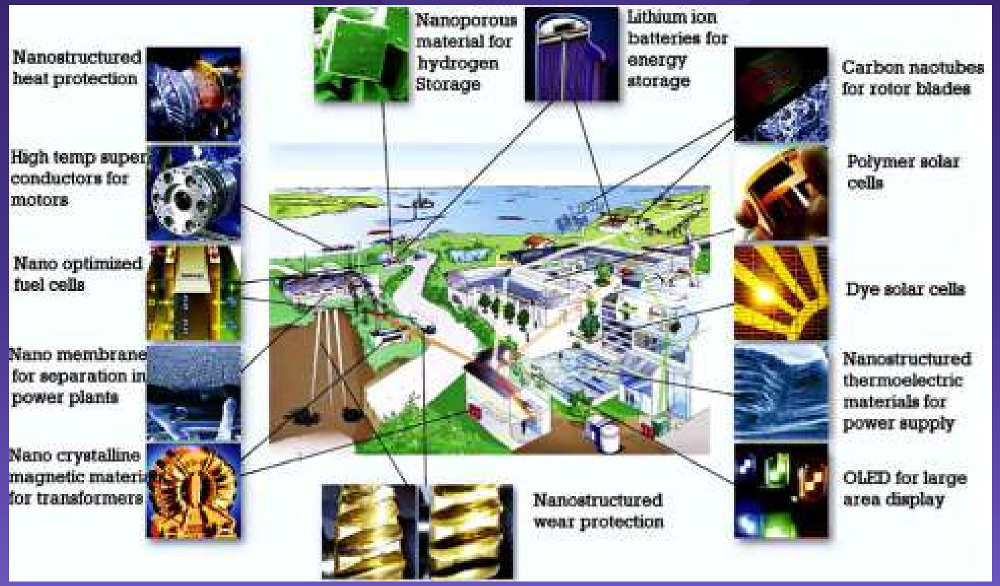
ELECTRONICS AND TECHNOLOGY



- Semiconductors and Quantum Dots
  - Nanoparticles exhibit quantum effects that enhance electronic and optical properties.
- Sensors
  - Nanoscale materials are used to develop highly sensitive sensors due to their thermal and electrical conductivity.
- Nanowires and Nanotubes
  - These are used in advanced electronics, energy systems, and miniaturized devices.

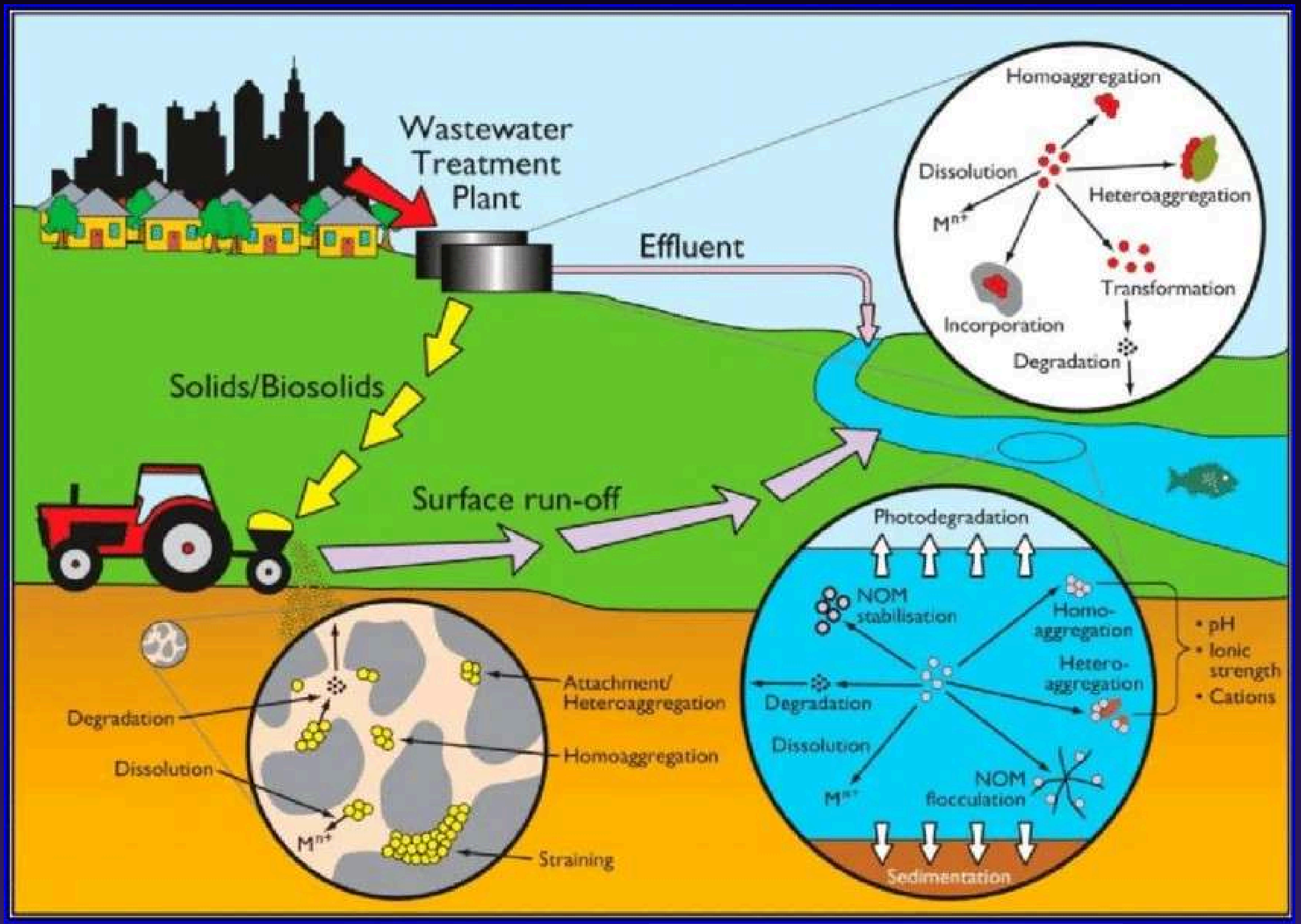
INDUSTRIAL APPLICATIONS

- Catalysis
  - Nanomaterials such as zeolites are used as catalysts in petroleum refining and chemical production.
- Detergents and Cleaning Agents
  - Zeolites are widely used in detergents to remove hardness ions from water.
- Fuel Production and Petrochemicals
  - Nanocatalysts are used in cracking and refining processes to improve fuel quality.
- Superplastic Materials
  - Nanostructured materials are used to produce lightweight and strong components, especially in aeronautics.



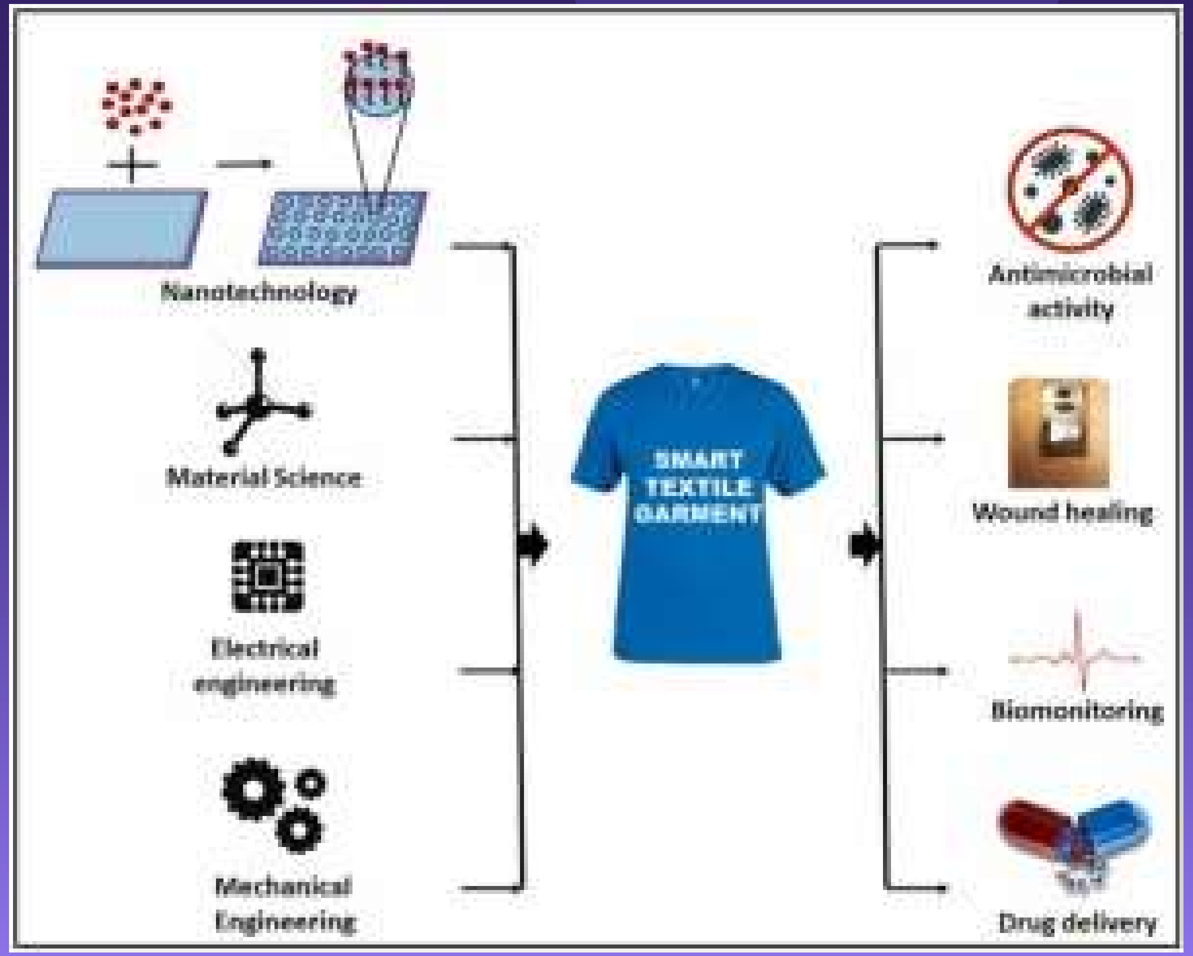
ENVIRONMENTAL APPLICATIONS

- Water Treatment
  - Nanomaterials like zeolites are used to remove contaminants and purify water.
- Pollution Control
  - Nanocatalysts are used in engines (e.g., diesel catalysts) to reduce harmful emissions.
- Waste Management
  - Nanomaterials are used in processing nuclear waste and environmental cleanup.



TEXTILE AND MATERIAL SCIENCE

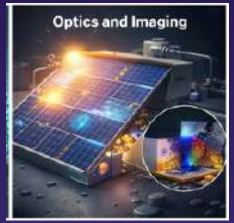
- Waterproof and Tear-Resistant Fabrics
  - Nanotechnology is used to produce durable and protective textiles.
- Nanocomposites
  - Improve strength, thermal stability, and fire resistance of materials.
- Improved Mechanical Properties
  - Nanomaterials enhance elasticity, durability, and resistance to deformation.



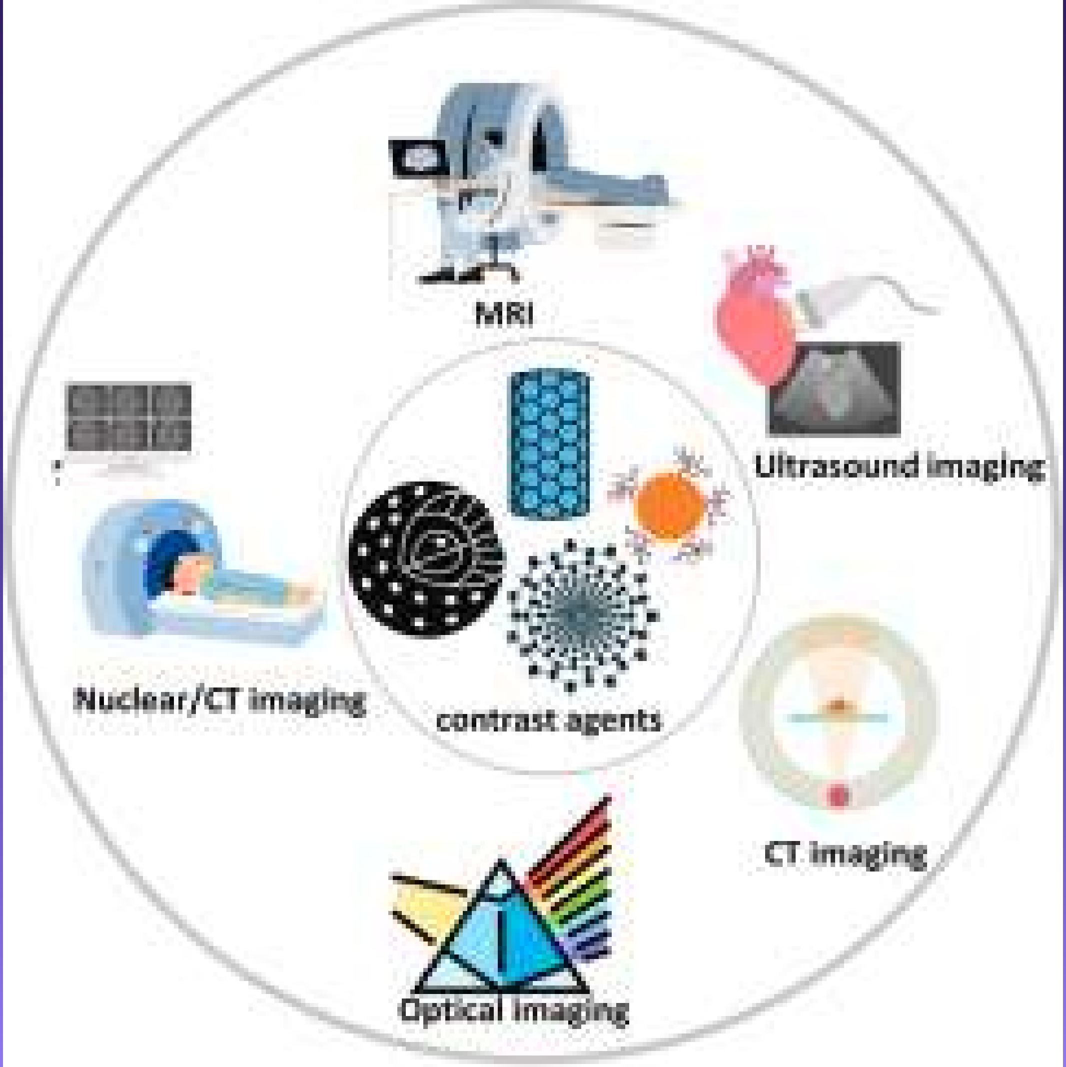
ENERGY APPLICATIONS

- Solar Energy Absorption
  - Nanoparticles improve absorption of solar radiation in solar cells.
- Energy Storage and Conversion
  - Nanomaterials are used in advanced batteries and energy devices.

OPTICS AND IMAGING



- Phosphors in Displays
  - Used in TV screens, fluorescent lamps, and plasma displays.
- Anti-counterfeiting Technology
  - Nanomaterials are used in banknotes and security tagging.
- Raman Scattering (SERS)
  - Enhances detection of molecules for chemical analysis.



RELATED RESEARCH

Fruit Research

REVIEW

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https://doi.org/10.48130/frures-0024-0037

Fruit Research **2025**, 5: e004

Applications of nanomaterials in agricultural production

Yali Zhang, Sang Lv, Haoyu Wang, Mingxing Tu, Zhumei Xi* and Xianhang Wang*

College of Enology, College of Food Science and Engineering, Viti-viniculture Engineering Technology Center of State Forestry and Grassland Administration, Shaanxi Engineering Research Center for Viti-Viniculture, Heyang Viti-Viniculture Station, Northwest A&F University, Yangling, Shaanxi 712100, China

* Corresponding authors, E-mail: xizhumei@nwafu.edu.cn; wangxianhang@nwafu.edu.cn

Abstract

Nanotechnology has been widely applied in the field of agriculture to meet the requirements of green agricultural development. In agricultural production applications, nanomaterials have been applied as nano-fertilizers, nano-pesticides, and nano-sensors. This article provides a detailed review of recent agricultural applications of nanomaterials. It will also focus on specific agricultural applications, such as transgenics, product preservation, stress resistance, and growth and development. Finally, the challenges of applying nanomaterials for agricultural research are summarized, and solutions are proposed to promote the safe and efficient utilization of nanomaterials in agricultural production to achieve sustainable agricultural development.

Citation: Zhang Y, Lv S, Wang H, Tu M, Xi Z, et al. 2025. Applications of nanomaterials in agricultural production. *Fruit Research* 5: e004 https://doi.org/10.48130/frures-0024-0037

RELATED COMPETENCIES IN K-12 CURRICULUM

GRADE 7

S7MT-Ig-h-5: Recognize that substances are classified into elements and compounds
S7MT-Ie-f-4: Distinguish mixtures from substances based on a set of properties

GRADE 8

S8MT-IIIc-d-9 Explain physical changes in terms of the arrangement and motion of atoms and molecules;
S8MT-IIIi-j-12 : Use the periodic table to predict the chemical behavior of an element.

RELATED COMPETENCIES IN K-12 CURRICULUM

GRADE 9

S9MT-IIe-f-16: Explain how ions are formed;

S9MT-IIg-17: Explain how the structure of the carbon atom affects the type of bonds it forms

S9MT-IIh-18 : Recognize the general classes and uses of organic compounds;

GRADE 10

S9MT-IIj-20: Investigate the relationship between: 1 volume and pressure at constant temperature of a gas 2 volume and temperature at constant pressure of a gas 3 explains these relationships using the kinetic molecular theory

REFERENCES

Saliha, A. I. B. (n.d.). Introduction to nanochemistry. AIB Saliha Publications.

Nanomaterials and nanochemistry. (n.d.).

College of Enology, Food Science and Engineering, Viti-viniculture Engineering Technology Center, Northwest A&F University. (2025). Fruit research. Fruit Research, 5, Article e004. https://doi.org/10.48130/frures-0024-0037